The Impact of Collaborative Scaffolding in Educational Video Games on the Collaborative Support Skills of Middle School Students

by

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To my supportive parents

Biographical Sketch

The author was born in Riga, Latvia. She attended Wellesley College and graduated with a Bachelor of Arts degree in mathematics and computer science. She began doctoral studies in computer science at the University of Rochester in 2010. After receiving her Master of Science degree from the university in October of 2011, she went on to pursue research in human-computer interaction under the direction of Henry Kautz and Christopher A. Egert.

The following publications were a result of work conducted during doctoral study:

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[3] A. Loparev, W. S. Lasecki, K. I. Murray and J. P. Bigham. Introducing Shared Character Control to Existing Video Games. In *Proceedings of the 9th International Conference on the Foundations of Digital Games*. Society for the Advancement of the Science of Digital Games, April 2014.

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Abstract

Collaboration is crucial to everything from product development in the workplace to research design in academia, yet there is no consensus on best practice when it comes to teaching collaborative skills. We explore one promising option: collaborative scaffolding in educational video games. Through this methodology, we can impart collaborative skills alongside topics like math and science in a medium already proven to be beneficial.

Before approaching our study design, we had to reconcile the large and often contradictory volume of literature on collaboration's definition and elements. From our review of over 25 studies and surveys, we developed a multilevel taxonomy with four main components: group quality, coordination, communication, and support. Researchers can utilize the taxonomy to isolate elements of collaboration relevant to their interests, and educators can use it to help identify collaborative skills that students lack. It can even act as a design guide for educational game developers.

For our own work, we chose to focus on scaffolding two subcomponents of support: direct and indirect helping. We ran mixed-factorial trials with over 70 local middle school students to observe behaviors and attitudes before, during, and after collaborative video game play. To gather data, facilitate play, and present scaffolding to students, we made several modifications to an existing educational physics game, including the creation of a scaffolding architecture and the implementation of a data collection infrastructure. Researchers and game developers can build on our designs and learn from the challenges we

encountered to help develop and evaluate future scaffolding systems.

During trials, we recorded audio, made direct observations, and logged in-game events. From our data, we found that collaborative scaffolding had no impact on helping behaviors, but, more interestingly, had a negative impact on motivation and attitudes toward collaboration. These results highlight the need for further exploration into behavioral effects and the external factors that can impact the value of this promising approach.

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1 Introduction

As a 21st century skill [81], collaboration is a necessary building block of success, but researchers and educators have yet to uncover the best approach to cultivating collaborative abilities. We propose the novel methodology of intelligent collaborative scaffolding feedback in educational video games. From the commercial [11, 108] sector to the research [83, 91] sector, video games have already been used to teach concepts like reading [82], math [57], and science [101] with proven benefits [54, 65, 35], especially for games that promote collaboration [6, 78, 79]. The addition of specialized scaffolding allows us to leverage these inherent advantages while facilitating the parallel teaching of collaboration with other topics.

To identify content to scaffold, we reconciled the collaboration research space through an analysis of over 25 studies and surveys from a wide range of disciplines and with diverse target demographics [62]. The resultant multilevel taxonomy has four components, each consisting of a series of more detailed subcomponents. We summarize the main components below:

- *Group Quality* - Aspects indirectly related to task completion that define the general feel of the group and its dynamics.
- *Coordination* - The organization of group members and the synchronization of actions through strategy creation, understanding, and execution.

- *Communication* - The exchange of facts, thoughts, and opinions.
- *Support* - The aid group members provide with respect to task completion, role execution, and concept understanding.

Our work centers on two subcomponents of support: performing part of a task for someone (direct helping) and providing guidance (indirect helping). In the context of collaboration, such actions are considered help only if they are welcome. Dismissing an individual and taking over without consent does not constitute helping behaviors, nor does giving hints to a person who wants to solve a problem without external aid.

After modifying an existing educational video game, we ran trials at Midlakes Middle School and the Rochester Institute of Technology's Kids on Campus summer program. Our results bring academics and educators one step closer to understanding and refining the way we integrate 21st century skills into today's classrooms. In addition, researchers in the fields of Computer Supported Cooperative Work (CSCW) and Computer Supported Collaborative Learning (CSCL) can utilize these results to inform future research directions and evaluation techniques.

Overall, our work provides five major contributions to various research fields related to collaboration, education, and technology. We developed a definition and a taxonomy that can adequately be applied to the context of collaborative scaffolding in educational video games, allowing researchers to explore the topic in more detail through an appropriate lens. We also implemented an infrastructure for collaborative scaffolding cues and data collection. The designs we created and our methods of addressing the challenges in developing this software can be used by future researchers and developers in the construction of their own technology. Lastly, in this dissertation we provide a discussion concerning our data on the effects of collaborative scaffolding, which can be used to define new research directions and iterate on the collaborative scaffolding design paradigm.

1.1 Relevance and Rationale

Because collaboration is an important 21st century skill, it is beneficial to prepare people at a young age for collaborative work. However, effective education cannot commence until we as a society agree on the best approach. One option is to give students opportunities to practice collaboration [30, 115] by instructing them to work together, varying reward contingencies, and constructing tasks that are interdependent [10]. This methodology has many variations. For example, instructors can assign roles [43, 46] and, like in Aronson's Jigsaw method [1], give different information to each person. They can also chose to monitor, scaffold, and regulate interactions [26]. These tactics are the most efficient with respect to time because collaboration can be addressed alongside other topics. An alternative approach is to create lessons focused exclusively on skills related to collaboration and communication [115], including discussion and helping [30]. Such concentration allows collaborative skills to be taught independent of domain to prevent frustration from attempting to master too much material at once.

No matter the method, the goal is to give students the opportunity to learn about and experience the collaborative process and collaborative interactions, which are especially advantageous for those of lower ability [30, 99]. Because collaboration can promote collaborative tendencies [96], it can potentially be used to alter the behaviors of people with individualistic inclinations who place more value on personal pursuits than group goals. Other behaviors that need to be overcome include minimizing help requests so as not to look "dumb" [73], as well as deliberately (free riding) or subconsciously (social loafing) putting in less effort so as to benefit from collaboration with limited contribution [113].

By aiming for greedy pursuits, group members minimize their contributions to group work and can incite the greedy behavior of others even outside of the group itself [38]. We need a deep understanding of collaboration and how to teach collaborative skills so that

we can combat selfish behaviors that negatively impact productivity and the experiences of fellow group members. To this end, we developed a multilevel taxonomy based on over 25 existing studies and surveys from a wide range of disciplines and with diverse target demographics. Researchers can identify aspects of collaboration relevant to their interests and at the appropriate scope, while educators can use the taxonomy to assess collaborative skill proficiency and adapt their curricula accordingly.

We used this taxonomy to guide the creation and assessment of an intelligent collaborative scaffolding feedback system in an educational video game. The results of our trials bring academics and educators one step closer to seamlessly integrating 21st century skill education into classroom curricula, and they contribute to the validation of video games both as an area of study and as an educational tool.

1.2 Objectives

To inform our research topic, we developed four research questions that vary on two dyads: help category and setting. We aim to explore instances of direct and indirect helping in both a video game and a classroom context. These focal points result in the following questions:

Question 1: Does collaborative scaffolding in an educational game promote more instances of performing part of a task for a group member (direct helping) than non-collaborative neutral scaffolding in the context of the game?

Question 2: Does collaborative scaffolding in an educational game promote more instances of providing guidance to a group member (indirect helping) than non-collaborative neutral scaffolding in the context of the game?

Question 3: Does collaborative scaffolding in an educational game promote more instances of performing part of a task for a group member (direct helping)

than non-collaborative neutral scaffolding in the context of a collaborative classroom activity?

Question 4: Does collaborative scaffolding in an educational game promote more instances of providing guidance to a group member (indirect helping) than non-collaborative neutral scaffolding in the context of a collaborative classroom activity?

1.3 Terminology

1.3.1 21st Century Skills

In 2002, the Partnership for 21st Century Skills [81] was founded and began work on a framework to outline capabilities crucial to success in the modern world. They separated these 21st century skills into three distinct categories, each filled with abilities that go beyond traditional academic topics and include such general competences as collaboration, as well as social and cross-cultural skills.

The first category is *learning and innovation skills*. These skills include *creativity and innovation* skills such as the ability to think creatively, work creatively with others, and implement innovations. *Critical thinking and problem solving* also falls under this category and includes the ability to reason effectively, use systems thinking, solve problems, and make judgments and decisions. *Communication and collaboration*, another subcategory, includes the ability to communicate clearly and collaborate with others.

Information, media and technology skills are those needed to interact effectively in today's computer-mediated world. Through *information literacy*, students have the ability to access, evaluate, use, and manage information, while *media literacy* includes the ability to analyze media and create media products. In turn, *information and communication*

technology (ICT) literacy encompasses the ability to apply technology effectively.

The last category of 21st Century Skills is *life and career skills*, which encompasses concepts like *flexibility and adaptability*. One set of skills in this category is *initiative and self-direction*, such as the ability to manage goals and time, work independently, and be a self-directed learner. *Social and cross-cultural skills*, another subcategory, include the ability to interact effectively with others and work effectively in diverse teams. In addition, *productivity and accountability* manifests through the ability to manage projects and produce results, while *leadership and responsibility* includes the ability to guide and lead others as well as to be responsible to others.

1.3.2 Collaboration

Collaboration is closely related to *cooperation* and *teamwork*. Some researchers make an effort to distinguish between these terms [2, 4, 19, 21, 32, 70, 77, 80, 92, 103, 120], while others use them interchangeably [6, 20, 41]. The popularity of each term depends largely on the area of study; computer science favors *collaboration*, most prominently in the areas of Computer Supported Collaborative Learning (CSCL) [19, 20, 21, 103] and Computer Supported Cooperative Work (CSCW) [4], while *cooperation* dominates the field of psychology [6, 24, 79, 93, 96, 113] and *teamwork* is featured in the industry [2, 84, 109] and military [67, 80] sectors.

To compound the confusion, collaboration is not well defined in literature. Some researchers define it by its behaviors and skills, which, in turn, are often defined themselves. Examples include:

- "cognitive and interpersonal skills needed to manage the flight within an organized aviation system" [98]
- "[knowledge, skills, and attitudes (KSAs) that] facilitate coordinated, adaptive per-

formance and support one's teammates, objectives, and mission" [2]

- "observable, non-technical behaviours that contribute to superior or substandard performance within a work environment" [31]

- "activities performed by team members in such a manner that each activity is coordinated with every other one and contributes to the superordinate goals of the unit or supports the activities of other members." [77]

Although a focus on these behaviors and skills can aid in understanding, defining a concept by its parts ignores external factors, such as context and motivation, and implies that collaboration does not extend beyond the listed components. Such definitions are too specific, leaving out *why* elements are numerous and relevant. Collaboration's nature can yield a more accurate definition than a description of its measurable effects, which is why some researchers instead view it as an activity, such as:

- "the mutual engagement of participants in a coordinated effort to solve a problem together" [92]

- "coordinated, synchronous activity that is the result of a continued attempt to construct and maintain a shared conception of a problem"; "a process by which individuals negotiate and share meanings relevant to the problem-solving task at hand" [107]

- "the willful contribution of personal effort to the completion of interdependent jobs" [113]

Others see collaboration as a context, with definitions like:

- "In a cooperative social situation the goals for the individual or sub-units in the situation under consideration have the following characteristics: the goal regions for

each of the individuals or sub-units in the situation are defined so that a goal-region can be entered (to some degree) by any given individual only if all the individuals or sub-units under consideration can also enter their respective goal-regions (to some degree)." [24]

- "A cooperative situation is described as one in which the goals of the participants are related in such a way that each subject will only achieve his goal if the rest invariably achieve their own." [34]

- "In cooperative group environments the goals of the separate individuals become so linked together that there is a positive correlation between their goal attainments" [96]

To reconcile the research space, we developed the following definition based on those listed above, borrowing heavily from Roschelle and Teasley [92] and Ryan and Wheeler [96]:

> *Collaboration* is the mutual engagement of participants, whose goals are interdependent and positively correlated, in a coordinated effort to complete a task.

As our first major contribution, this definition can be used by researchers, educators, and software developers to design learning objectives and evaluation criteria concerning collaborative scaffolding in educational video games.

1.3.3 Helping Behaviors

Although we use the terminology *helping behavior*, our definition is based on two closely related concepts: *helping behavior* and *backup behavior*. According to Porter et al. [84],

backup behavior occurs when help is required, while *helping behavior* occurs when help is unnecessary but may be advantageous. In contrast, Marks et al. [64] base their definition of backup behavior on Dickinson and McIntyre's [25] work and define it as occurring when "(1) providing a teammate verbal feedback behavior or coaching, (2) helping a teammate behaviorally in carrying out actions, or (3) assuming and completing a task for a teammate." Although Marks et al. call these backup behaviors, they also qualify as helping behaviors under Porter et al.'s definitions. Our definition of helping behavior aligns with Porter et al.'s, and we use two of Marks et al.'s backup behaviors as a focal point: performing part of a task for someone (assuming and completing a task for a teammate) and providing guidance (providing a teammate coaching).

1.3.4 Scaffolding

Wood et al. [116] coined the term *scaffolding*, meaning:

> [A] process that enables a child or novice to solve a problem, carry out a task or achieve a goal which would be beyond his unassisted efforts. This scaffolding consists essentially of the adult "controlling" those elements of the task that are initially beyond the learner's capacity, thus permitting him to concentrate up on and complete only those elements that are within his range of competence.

Our intelligent collaborative scaffolding feedback system uses prompts to supply players with guidance on when and how to effectively collaborate with statements like "I think my partner is going to need my help, too." and "Maybe we can make an easier path up if we work together." These phrases are positive, in line with scaffolding's formal definition.

However, scaffolding can also be neutral or even negative. In the context of an educational video game, *neutral collaborative scaffolding* (or just *neutral scaffolding*) includes statements that neither help nor hinder players, such as "Water tastes funny." and "I like

clouds.", while *negative collaborative scaffolding* hinders skill development with statements such as "You don't need anyone's help." and "Working with others is a waste of time." For our work, participants in the treatment group encountered positive collaborative scaffolding, while those in the control saw neutral scaffolding.

2 Background

Concepts relevant to our research, such as scaffolding, games, collaboration, and education, have been explored across many domains. They shaped our research questions and guided our study design. Below we outline these influences and their significance to our work.

2.1 Scaffolding

Wood et al. coined the term *scaffolding*, meaning a "process that enables a child or novice to solve a problem, carry out a task or achieve a goal which would be beyond his unassisted efforts" [116]. They went on to identify six functions of scaffolding, each of which we addressed in the following manner:

- *Recruitment* - Spurring the child's interest and directing the child's focus on the task. Our game entices players with content.

- *Reduction in degrees of freedom* - Limiting the number of actions a child can perform, the goal being to simplify the task. Our game limits player input channels naturally via technology restrictions and artificially through a narrow set of key mappings to in-game actions.

- *Direct maintenance* - Keeping the child motivated and continually progressing. Our scaffolding hints when collaboration is beneficial for progression.

- *Marking critical features* - Pointing out important items. Our scaffolding directs player attention when collaboration becomes a necessity.
- *Frustration control* - Preventing the child from becoming too stressed. Our scaffolding reminds players of the benefits of collaboration when they are stumped.
- *Demonstration* - Showing example solutions. Our game contains a video at the start showing how the basic mechanic works and benefits all players.

Effective scaffolding often incorporates the concept of *fading*: "once the learner has a gasp of the target skill, the master reduces his participation (fades), providing only limited hints, refinements, and feedback to the learner, who practices by successively approximating smooth execution of the whole skill" [22]. To facilitate fading, our software tracks player input and in-game events to identify situations in which collaboration is advantageous but overlooked. By only providing feedback when necessary, scaffolding diminishes as players learn to identify collaboration opportunities on their own.

According to Quintana et al. [86], aside from fading, scaffolding should also:

Guideline 1: Use representations and language that bridges learners' understanding

Guideline 2: Organize tools and artifacts around the semantics of the discipline

Guideline 3: Use representations that learners can inspect in different ways to reveal important properties of underlying data

Guideline 4: Provide structure for complex tasks and functionality

Guideline 5: Embed expert guidance about scientific practices

Guideline 6: Automatically handle nonsalient, routine tasks

Guideline 7: Facilitate ongoing articulation and reflection during the investigation

Quintana et al. presented various scaffolding strategies for each of these guidelines, mostly pertaining specifically to scientific scaffolding. The few strategies relevant to our own work, and which we considered when developing our content, are outlined below:

1c: Embed expert guidance to help learners use and apply science content

2a: Make disciplinary strategies explicit in learners' interactions with the tool

5a: Embed expert guidance to clarify characteristics of scientific practices

5b: Embed expert guidance to indicate the rationales for scientific practices

Given that traditional scaffolding systems tend to address traditional educational topics like math and reading, Quintana et al.'s focus on scientific scaffolding is unsurprising. That is not to say that research has not been done into collaborative scaffolding in general [121] and topics related to collaboration, such as exploratory talk [75, 90] and reaching group consensus [74]. However, little of it addresses helping behaviors in particular, which is the focus of our work.

A concept closely related to scaffolding is Vygotsky's *zone of proximal development*: "the distance between the actual developmental level as determined by independent problem solving and the level of potential development as determined through problem solving under adult guidance or in collaboration with more capable peers" [112]. Through the lens of scaffolding, it is "the distance between problem solving abilities exhibited by a learner working alone and that learner's problem solving abilities when assisted by or collaborating with more experienced people" [60]. It functions as a measure of a scaffolding system's effectiveness; the larger the zone, the more useful the approach. A zone of minute size indicates the system is ineffectual because knowledge acquisition is the same regardless of its presence. In the context of formal scaffolding system evaluations like ours, significantly positive improvements in treatment groups when compared to control groups indicate large

zones of proximal development, while insignificant but still positive differences suggest small ones.

2.2 Collaboration among children

Studies show that young girls perform better in a collaborative environment than a competitive one, while boys perform equally well in both [36, 44]. Thus, a collaborative context is beneficial to everyone, while a competitive one is detrimental to half of the child demographic. Facilitating an effective collaborative environment requires child proficiency in collaborative skills, which can be cultivated through the type of learning and reinforcement our scaffolding system aims to provide.

Just as collaborative skills can affect collaborative environments, collaborative environments can affect predispositions toward collaboration. Ryan and Wheeler [96] found that fifth and sixth grade students given collaborative, rather than competitive, lessons exhibited more collaborative tendencies during post-lesson game play. These tendencies came in four categories: seeking help for one another, positive responses to help sought, volunteering help for others, and establishing group strategies for sharing resources with individuals in need. Our work also centers on helping skill retention, although we distinguish help categories by help methodology instead of initiation. Moreover, rather than compare the effects of a collaborative vs. a competitive task, we concentrate on collaborative vs. neutral scaffolding during in-game collaborative missions.

Thus far, we have discussed collaboration among children within a group, but we can also consider collaboration between groups. Puntambekar et al.[85] had seventh graders work together to design a robot. They found that during intragroup collaboration, students concentrated on structure and behavior, while between groups they spend more time discussing functionality. In our research, we focus on intragroup interactions, but acknowl-

edge that results may not be the same with intergroup collaboration. We leave a closer inspection of intergroup helping skills for future work.

2.3 Collaborative learning through games

Researchers have touted the benefits of games as an educational tool [54, 65, 35] that improves self-efficacy and self-esteem [83], as well as increases motivation to learn [91]. Through collaborative play in particular, children have shown an increase in collaborative behaviors [6, 78] like sharing [79] and a decrease in aggression [6].

Often such findings come from collaboration interventions. For example, Blazik [8] showed that collaborative play improves positive attitudes toward peers, increases the number of peers that students feel they can work with, and makes students and teachers feel like the class is getting something positive out of collaborative interactions during the intervention. Landazabal [59] also performed an intervention and found that playing games that encourage collaboration and sharing improves assertive behavior and self-concept, while Garaigordobil et al.'s [34] collaboration intervention revealed that it facilitates better leadership skills, more leaders, and higher intra-group acceptance while decreasing peer rejection and increasing giving/taking, asking/receiving, helping actions, and productivity during a collaborative activity.

2.3.1 Collaborative learning

There are four types of classroom goal structures: cooperative, competitive, individualistic, and no structure [47]. Our interest lies in the cooperative variety, which has the following four properties [49]:

- *Positive interdependence* - Interdependence of group member goals, tasks, resources, roles, and rewards
- *Individual accountability* - Material is learned by each group member
- *Face-to-face interaction* - Prevalent conversation and non-verbal communication among students
- *Cooperative skills* - Appropriate use of social skills needed for collaboration

For collaboration, a cooperative environment can have many benefits over a competitive one. For example, Deutsch [24] found that in an undergraduate classroom setting, cooperative groups are better at coordination, have more diverse member contributions, and overall achieve higher productivity. Johnson and Johnson [47] explored a diverse set of literature on cooperative classrooms and found that, when compared to a competitive structure, a cooperative structure leads to better performance, as well as more positive attitudes toward topics, tasks, and other people. The authors concluded that a cooperative structure is best when it comes to problem solving tasks, which are common both in games and in the classroom. In a later review of the literature [48], the same authors identified additional benefits like the encouragement of inter-student support, the motivation of academic interests, and the facilitation of 21st century skill acquisition. By promoting a cooperative structure through our scaffolding content, we aim to increase the number of students who benefit from these advantages.

2.3.2 Collaborative video games

Playing collaborative video games, even those not strictly educational, has been shown to yield positive results. Inkpen et al. [44] revealed that video game play together is more productive than play in isolation and, for female-only pairs, is also more productive than

playing side-by-side on two computers instead of one. In addition, they asserted that having a partner could be a cause of increased motivation to continue play.

To give people these benefits, collaborative video games have to entice player participation through effective design choices. According to Rocha et al. [89], successful collaborative games have elements like complementary roles, synergies between abilities, and shared goals. El-Nasr et al. [102], who expanded on Rocha et al.'s work, found additional patterns, including shared characters, shared puzzles, and limited resources. Zagal et al. [117] performed an independent in-depth investigation and developed a set of lessons and pitfalls for designing collaborative games. The lessons mainly addressed the crucial role of selfishness and autonomy, while pitfalls centered on the importance of incentives for collaboration, success, and repeated play. When modifying our video game, we used these design patters, lessons, and pitfalls to guide development.

2.4 Theoretical influences

Collaborative play can be deconstructed into two components: play and collaboration. By understanding the form and functions of these processes, we gain a stronger grasp of scaffolding's role in collaborative video games and its impact on player experience.

2.4.1 Definitions and properties of play

According to Huizinga in his seminal book on play, "play is a voluntary activity or occupation executed within certain fixed limits of time and place, according to rules freely accepted but absolutely binding, having its aim in itself and accompanied by a feeling of tension, joy, and the consciousness that it is 'different' from 'ordinary life"' [42]. Caillois [16] expanded on Huizinga's work by characterizing play as voluntary, within a particular time and space, having an unforeseen conclusion, self-contained in its creations, governed

by rules, and evoking a degree of make-believe.

In addition, he classified games into four categories:

- *agôn* - competitive games based on a single quality with winners and losers
- *alea* - games based purely on chance
- *mimicry* - play centered around pretending to be someone or something else
- *ilinx* - play which centers around creating a distortion of perception

We argue that collaborative games best fit into agôn, where players compete against their environment. Sutton-Smith [106] also classified play by its forms, mostly of the informal variety like playful behaviors and vicarious audience play. Although he did not explicitly address collaborative games, board and card games were placed under contests, next to gambling, sports, and pool. In both classifications there is no adequate fit for collaborative games without resorting to characterizing such play as a competition between the set of players and their environment, placing emphasis in the wrong place. Collaboration is not about conflict, but rather banding together.

Sutton-Smith also notably stated that as children get older, "they play increasingly complex games and develop complexity in the social skills required to play." If we apply this notion to collaborative games, then we can say that such games help children develop the collaborative skills required to play, which may translate to non-game settings, such as group projects for school.

No matter the complexity of the game, an important aspect of play is its setting, often called a *magic circle*. Huizinga coined this term and defined magic circles as "forbidden spots, isolated, hedged round, hallowed, withing which special rules obtain" [42]. He went on to explain that they are "temporary worlds within the ordinary world, dedicated to the performance of an act apart." Someone who violates or ignores the rules is a "spoil-sport."

A spoil-sport destroys the play world, and therefore receives a harsher punishment than a "cheat," who still acknowledges the magic circle. In traditional games, the magic circle is enforced by the players themselves, while in video games the enforcement is done by the game. In both cases, players suspend the rules of the real world and immerse themselves in an alternate setting.

During play within a magic circle, participants, whether consciously or subconsciously, strive to enter a state of flow, a term coined by Csikszentmihalyi meaning "the state in which people are so involved in an activity that nothing else seems to matter; the experience itself is so enjoyable that people will do it even at great cost, for the sheer sake of doing it" [23]. In an optimal state, game difficulty is enough to avoid boredom, but not so great as to cause anxiety. In a state of boredom, the player quits from lack of interest or looks for additional challenges. In video and board games today, these challenges my take the form of playing on a harder difficulty or finishing under a time limit. If the game causes anxiety, the player either quits out of frustration or works harder to improve and overcome challenges. With educational games, it is important to gauge how much a player knows, so that excessively difficult educational challenges do not cause the player to quit. Support for flow in educational games with collaborative mechanics is even trickier because the level of player skill can differ across group members. In addition, the experience of playing is not only impacted by the game itself, but also by the social interactions of the players.

2.4.2 The collaborative process

Although our work views collaboration as an activity with a set of functions, other researchers have treated it as a process. According to Gray's three phase framework [37], at first a problem is defined, available resources are identified, relevant stakeholders are recruited, and commitments to the collaboration are made. This phase is called *problem*

setting. Next, during the *direction setting* phase, ground rules are established and strategies are developed. The collaborative process ends with *implementation* as group members execute strategies under careful monitoring.

Other researchers have divided the process into more precise steps. For example, Kagan [51] identified six stages, which essentially expand on Gray's three phases in the following manner:

Stage 1: *Formation* - Like in Gray's *problem setting*, a problem is identified, people chose to join a group, and the viability of collaboration is evaluated.

Stage 2: *Conceptualization* - Objectives are defined and roles and responsibilities are assigned, making this stage signify the start of Gray's *direction setting*.

Stage 3: *Development* - Strategies are developed, making this stage a continuation of Gray's *direction setting*.

Stage 4: *Implementation* - Strategies are executed. Like the name suggests, it mirrors the start of Gray's *implementation* phase.

Stage 5: *Evaluation* - Group self-assessment occurs, making this stage a continuation of Gray's *implementation* phase.

Stage 6: *Termination* - Collaboration ends, and so does Gray's *implementation* phase.

The helping behaviors that we focus on in our work occur mainly during the implementation phase, when group members part to perform independent tasks.

2.5 Computer Supported Cooperative Work (CSCW)

Computer Supported Cooperative Work (CSCW) is "an endeavor to understand the nature and requirements of cooperative work with the objective of designing computer-based tech-

nologies for cooperative work arrangements" [100]. These technologies are often referred to as groupware: "computer-based systems that support groups of people engaged in a common task (or goal) and that provide an interface to a shared environment" [29]. Although the focus is on adults in the workplace rather than children in the classroom, CSCW includes research on how groups collaborate to perform problem-solving tasks, making work in this field relevant to our research.

2.5.1 Groupware Design

Building a system for groups instead of individuals brings new challenges to software design. Grudin [40] discussed eight such challenges related to the benefits of groupware, technological limitations, user experience, and adoption, four of which are directly relevant to collaborative educational video games. For example, contradictory literature and the qualitative nature of collaboration increase the concept's *difficulty of evaluation*, while time and resource constraints in the classroom hinder *the adoption process*. In addition, some games may require teacher input to customize content despite the *disparity in work and benefit*. For our work in particular, *exception handling* via scaffolding feedback has to account for a large number of misconceptions and is visible to all players despite not everyone needing it.

Such challenges lead to design issues that groupware developers need to address, including the following [29]:

- *What You See Is What I See (WYSIWIS) issues* - Interface consistency across users ensures a shared context, but prohibits customization. An alternative is relaxed WYSIWIS rules in which the display space, time of display, participants affected, and /or the congruence of view vary [104].

- *Group focus and distraction issues* - When and why a group member alters an ele-

ment should be clear. If not, sudden changes can be distracting.

- *Issues related to group dynamics* - Interfaces should reflect usage patterns.
- *Issues related to screen space management* - The number of windows shared by a group can be unwieldy, so it is important to have an effective window management scheme.
- *Issues related to group interface toolkits* - Because of how complex group tools can be compared to single-user software, a simple conversion from one to the other is often not enough. Instead, new features and rules must be added to avoid unnecessary clutter.

Collaborative games encounter some of these same issues, mainly the need to design the interface for a group rather than a single user, grab attention but not distract, take group dynamics into account, and efficiently utilize screen space.

2.5.2 Inter-Organizational Collaboration (IOR)

Ring and Van De Ven [88] described the collaborative process of inter-organizational relationships (IOR) as a repetitive sequence of stages. Joint expectations are developed or modified (*negotiations*) before plans and a contract are formed (*commitments*), and ultimately plans are implemented (*executions*). Although aimed at the organization level, we can imagine such stages on the personal level: group members join together to overcome a common challenge. They develop strategies to accomplish their goals, and then implement those strategies.

The process of IOR can also be viewed from the perspective of its dimensions [110]:

- *Governance* - Processes for jointly making decisions

- *Administration* - A structure of leadership roles and responsibilities
- *Organizational autonomy* - Conflict between the self and the group
- *Mutuality* - Interdependence through mutual benefits
- *Trust and reciprocity* - Showing good faith with the belief that other parties will do the same

These dimensions reflect the rigid impersonal nature of inter-company collaboration. Administration and governance may be weaker in a small group of children, but the rest are all still present in this alternate context.

2.6 Psychology

In the field of psychology, researchers have explored the advantages of collaboration during various styles of play. For example, Rosen [93] measured the effects of sociodramatic play, during which "several children take on different roles and interact with each other in terms of a situation that they have spontaneously created" [93]. This type of play requires collaboration in the form of following appropriate social interaction norms, and often involves collaborative scenarios, such as a team of scientists. The author showed that such play improves role-taking skills, group problem-solving behavior, and effectiveness in solving group problems requiring maximum cooperation and minimum competition.

Aside from collaboration, research on prosocial behavior is also prevalent in the psychology literature. Prosocial behavior is "voluntary, intentional behavior that results in benefits for another; the motive is unspecified and may be positive, negative, or both" [27]. Within the context of collaboration, prosocial behavior occurs when group members support each other emotionally or through aid in understanding a concept or completing a task. Examples include comforting, sharing, and helping [71].

Such prosocial behaviors can be improved through collaborative play. When compared to traditional games, collaborative games result in more sharing post-play [79]. Furthermore, during collaborative, rather than competitive, exercise play, children exhibit more instances of prosocial interactions [39].

Like collaboration, prosocial behavior can be deconstructed into components. Zahn-Waxler and Radke-Yarrow [118] categorized it into help, statements of sympathy, protection, physical comfort, sharing, and cooperation, while Brief and Motowidlo [14] focused on a workplace context and identified classifications centered on concepts like assistance, involvement, and loyalty. When developing our taxonomy of collaboration, we used these categories to guide our design.

Although these results are promising, of note is that cooperativeness is not universal. Johnson [45] found that an elementary school child's cooperativeness is related to skill in recognizing the feelings of others, understanding the causes for the feelings, and taking the overall affective perspective of others, while Wagner [113] proved collaborative tendencies are impacted by people's individualism-collectivism inclinations. *Individualism* refers to when collaboration "is motivated by the contingent satisfaction of personal interests" [114], and *collectivism* refers to when it "stems from the pursuit of interests shared among the members of a collectivity (even if these pursuits sometimes conflict with members' immediate personal desires)" [114]. Collectivists are more likely to exhibit collaborative behavior, while individualists have the opposite predisposition, so students with high levels of individualism stand to gain the most from collaborative scaffolding but are probably also the least likely to seek out collaborative games.

2.7 Implications

When performing our literature review, we noticed patterns in the description of collaboration and the aspects of collaboration researchers tended to focus on. From these patterns, we began to develop the multi-level taxonomy described in the next chapter, which in turn helped us refine our research direction.

The influence of this background work was not limited to only the next step in our research process. Past work in scaffolding helped us develop our scaffolding content and timing, while research in collaborative learning through games provided us with an overview of our research space. In addition, theoretical influences concerning play and the collaborative process allowed us to examine things from a more abstract perspective, while work in Computer Supported Cooperative Work and Psychology gave us a wider context in which to ground our research.

3 Taxonomy

Despite the large volume of research centered on collaboration, there exists no agreed upon terminology, definition, or deconstruction of the concept that adequately applies to collaborative scaffolding in educational video games. As an integral part of this dissertation, our second major contribution was the development of a multilevel taxonomy (Figure 3.1) based on an analysis of over 25 existing studies and surveys from a wide range of disciplines (Figure 3.2) and with diverse target demographics [62] (Table 3.1), which we used to reconcile the research space, as well as to guide our study design and evaluation methodology. Due to the ambiguity and interchangeability of various terms similar to collaboration, we included papers and articles pertaining to collaboration, cooperation, and teamwork (Figure 3.2), as well as those on prosocial behavior, a subset of collaboration focused on help and support. Because of our emphasis on collaborative problem solving, we chose to omit research on other collaborative scenarios, such as collaborative learning, unless it had broader implications.

3.1 Data Types

As part of our framework development, we considered potential assessment vectors for each subcomponent to guide our data collection. Because the importance of different categories of data and how they interrelate is not always clear, we reviewed the types of large

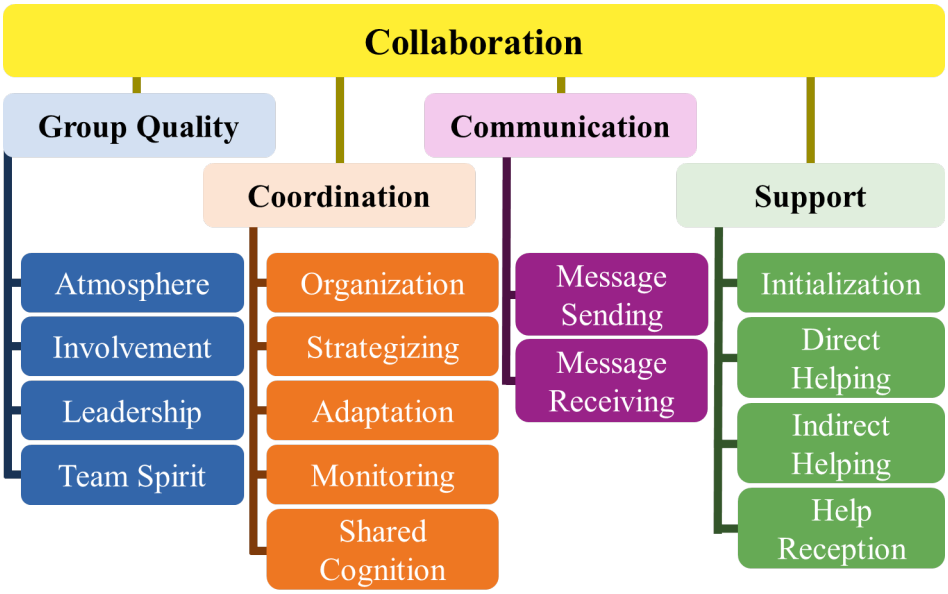


Figure 3.1: Taxonomy of collaboration visual overview



(a) Discipline distribution

(b) Terminology distribution

Figure 3.2: Literature review distribution

and diverse data sets that software can support. Below we outline the kinds of data we chose to use based on this examination, which we connect to the subcomponents of our taxonomy in the next section. Although aimed at the educational video game space, many are applicable to other domains.

The first type of data we identified is *audio/video*, which can be extracted from either a computer's built-in hardware or through external tools. Audio data takes the form of volume and tone levels, although we can use advanced software techniques to extrapolate

Table 3.1: Taxonomy of collaboration literary grounding

Component	Category	Referenced In
Group Quality	Atmosphere	[21] [24] [32] [67] [78] [93] [94] [103] [113]
	Involvement	[4] [20] [24] [32] [41] [68] [93] [103] [109] [120]
	Leadership	[2] [31] [32] [67] [68] [70] [80] [103] [109]
	Team Spirit	[2] [24] [32] [6] [67] [70] [78] [80] [94] [103]
Coordination	Organization	[24] [77] [93] [120]
	Strategizing	[2] [19] [20] [21] [24] [31] [32] [41] [68] [70] [77] [93] [94] [96] [109]
	Adaptation	[2] [24] [32] [56] [68] [70] [77] [94] [120]
	Monitoring	[20] [21] [31] [32] [41] [56] [67] [70] [77] [94] [103] [107] [109]
	Shared Cognition	[2] [20] [24] [31] [41] [56] [68] [92] [94] [103] [107] [109] [120]
Communication	Message Sending	[2] [4] [20] [21] [32] [67] [70] [77]
	Message Receiving	[24] [67] [70] [77] [103]
Support	Initiation	[6] [20] [32] [34] [41] [67] [70] [78] [79] [84] [94] [96]
	Direct Helping	[6] [70] [78] [94] [84]
	Indirect Helping	[6] [34] [67] [68] [70] [78] [79] [80] [84] [94] [109]
	Help Reception	[70] [80]

additional information, such as speaker-based segmentation and word identification. Video data consists of a set of pixels, potentially along with specialized information like depth and heat level. We can use computer vision to infer other facts, such as stance and facial expression. Although useful, audio and video data can be difficult to gather. Accurate collection can require expensive equipment, and may be impossible if suboptimal lighting conditions and excessive noise pollution are present.

The second type of data is based on player *input*, usually through either a keyboard and mouse combination or through gestures on a touchscreen. Keyboard data contains information about a key's value when it is pressed. For economic reasons, low- to mid-range keyboards are limited in how many keys can be recognized simultaneously. However, software developers have learned to work around this limitation when designing interactions, so it rarely has an impact. A keyboard is often paired with a mouse, whose position information, as well as mouse button and scroll wheel status, can be stored. While we can easily connect keyboard data to events through a one-to-one relation between key presses and their effects, interpreting mouse data involves understanding how mouse position relates to objects in the virtual space.

Aside from traditional computers with mice and keyboards, today's classrooms also occasionally employ tablets, which gather input through gestures and taps on the screen. Input data appears as a set of binary coordinates, each indicating if a touch occurs at that point. Usually we work at a slightly higher level: single points of contact based on interpreted finger positions. We can use gesture recognition techniques to even identify the actions a player attempts to perform. Because useful touchscreen data has to be interpreted, the information is less accurate than that of a keyboard and mouse.

The next set of data categories are types of events. *Progress events* mark incidents that indicate player progression, such as the introduction of a new goal or the completion of an activity step. *Scaffolding events* are those related to the actions of a piece of software's

scaffolding system. Often such data contains information concerning content, as well as whether it is an instance of appearance or disappearance.

Events occasionally take the form of interactions. The first type is *player/player interaction* and refers to actions done by players to players, such as giving or taking objects. In contrast, a *player/environment interaction* occurs when a player performs an action that does not directly involve others, such as lighting a torch or pushing a button. The third type is *environment/environment interaction*, which includes events that do not directly involve players. These often occur as a side effect of player/player or player/environment interactions, such as an object bouncing against a wall after a player threw it. Interaction data frequently takes the form of information about the objects and action involved, although details can vary depending on context.

Both events and interactions can affect a piece of software's two types of state. The *global state* is the current overall software state, such as the main menu, activity screen, or tutorial, while *local state* refers to the position and state of objects within the virtual environment, such as player location and whether a switch is turned on.

3.2 Components

The crux of our framework is the deconstruction of collaboration into four general components, each of which consists of a series of more detailed subcomponents. All components and subcomponents are grounded in existing literature, which we use to clarify and bolster our taxonomy. We also include examples of evaluation for each subcomponent, as summarized in Table 3.2, based on data types outlined in the previous section.

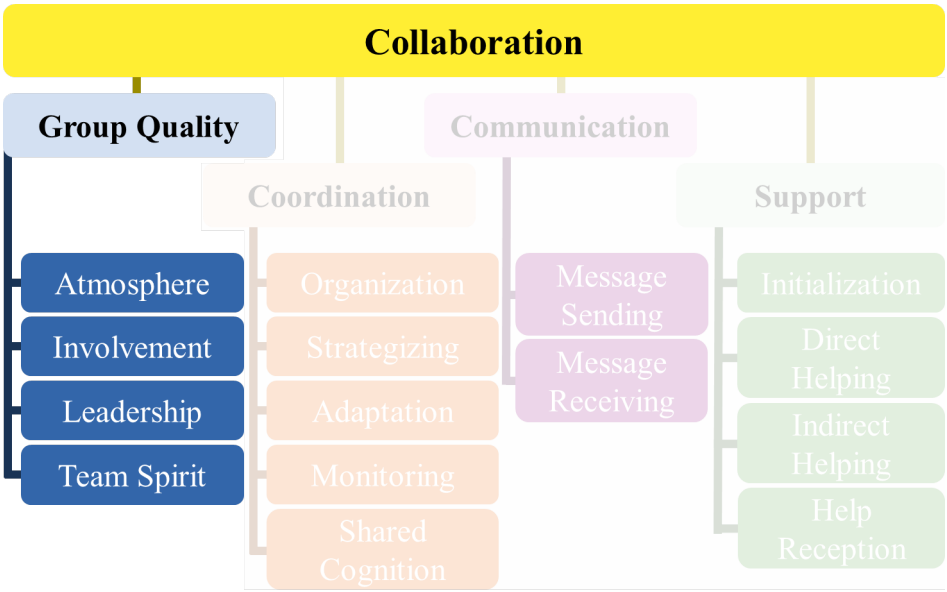


Figure 3.3: Group quality

3.2.1 Group Quality

The first component in our taxonomy is *group quality* (Figure 3.3): group aspects indirectly related to task completion. Even when each group member is highly intelligent and capable of performing all necessary functions, if members feel isolated, lack motivation, or simply do not get along, group performance can suffer. For example, when a conflict arises, ineffective conflict resolution can lead to a prolonged period of time dedicated to overcoming the dispute rather than completing the task. Because of its many facets, we split group quality into four subcomponents.

The first subcomponent is *atmosphere*, which can be viewed on various scales, such as control or hostility during group interactions [93]. Groups can improve atmosphere through positive relations among group members [24, 32, 67, 78], ample encouragement of communication [32], and effective conflict resolution [32, 94, 103]. Atmosphere is also affected by the value members place on personal gains versus those of the group [113] and by the attitudes of group members toward group participation [93], group work [21], and

Table 3.2: Scaffolding examples

Component	**Subcomponent**	**Example Positive Event**	**Example Scaffolding**
Group Quality	Atmosphere	Feelings of empathy and joy	"You may need to work together."
	Involvement	Player/environment interactions	"Everyone needs to do their part."
	Leadership	Assertive speech	"Think about taking charge."
	Team spirit	"That was awesome!"	"Every success is important."
Coordination	Organization	"You be the ... while I be the ..."	"Remember your responsibilities."
	Strategizing	Quick success	"You should come up with a plan."
	Adaptation	Quick success after context shift	"You should re-think your strategy."
	Monitoring	"You were...", "You didn't..."	"Think about why it's not working."
	Shared cognition	"I thought you meant..."	"Make sure you understand others."
Communication	Message sending	No recipient confusion	"Make your speech concise."
	Message receiving	Recipient re-uses terms	"Listen to what others have to say."
Support	Initiation	"Could you help me?"	"Don't be afraid to ask for help."
	Direct/Indirect helping	"Let me show you."	"Helping your group helps you."
	Help Reception	Pause to read scaffolding	"It's okay to ask for clarification."

the task at hand [67].

To evaluate atmosphere, we can infer attitudes through the affective analysis of audio data, video data, or a combination of the two [119]. By augmenting this approach with a keyword analysis focused on positive words, negative words, and words of encouragement, we can increase the accuracy of the assessment. To ensure an emotion stems from player/player interactions, we can suspend analysis during overly exciting events within the digital realm, such as goal completion or repeated failure of an intended action. From a glimpse into the distribution of shared resources in a software environment, we can also note negative hording tendencies, especially when sharing benefits others without consequence to the hoarder.

Once we identify a negative atmosphere, we can correct the behaviors involved through scaffolding. Feedback can take the form of direct statements of encouragement, such as "You may need to work together to solve this.", or it may be visual, demonstrating how players can interact in mutually beneficial ways within the digital world. Ideally, the scaffolding is implicit, for example through an abundance of direct collaboration opportunities.

Effort toward the development of a positive atmosphere [32] is an example of member *involvement*. Involvement can be thought of in a general sense [4, 32, 68, 93, 103, 120] or more specifically in relation to certain aspects of group work, including activities like group discussions [20]. It can take the form of involvement in group quality through the encouragement of others' involvement [32, 68], or in the effective undertaking of the task, such as through attention during group discussions [24], an interest in everyone's performance [41], and a strong focus on the task [109].

Using audio and video data, we can measure involvement through how often and for how long each player speaks, taking particular note of when it occurs after a question is asked. Through software, we can examine how often group members perform a public action, such as pressing a button or moving an object, relative to their peers. If a player is

silent and does not contribute to progression, we should introduce scaffolding, for example through statements such as “Everyone needs to do their part to reach the goal.” In addition, each player can be given a unique set of possible actions, making contribution mandatory for success. If a piece of software includes demonstrations or tutorials, all player avatars should be present, with extra time given to the avatars of uninvolved individuals.

To uphold a positive atmosphere, it is sometimes necessary for someone with *leadership* skills [68, 103] to contribute during group work. There may be a designated leader, as often occurs in the commercial and military sectors, or group members may share the responsibility of performing leader functions. Leadership skills include direction [2, 32, 67, 70, 80], positive atmosphere facilitation [2, 32], discussion mediation [103], performance assessment [2], and the effective organization of individuals and tasks [2, 67, 109], all while exuding authority [31] and assertiveness [31, 109].

To detect instances of leadership, we can identify key guiding phrases from audio data, such as “We should...”, “Let’s...”, and “Follow me to ...”, and use affective analysis to measure assertiveness. Leadership requires participation, so we should also incorporate our previous assessment of involvement. The easiest way to promote leadership is to create situations in the digital world that necessitate leadership skills.

Another leadership skill is providing emotional care [67], often in the form of raising *team spirit* [2, 24, 32]. It can originate from positive physical contact [6, 70, 78] or words of encouragement [94, 103], motivation [70, 80, 103], and praise [70, 80]. We distinguish emotional support from support related to a task, the latter of which is its own component of collaboration.

We can measure team spirit through key phrases in audio data, such as “You can do it.”, “It’s OK.”, and “That was awesome!” If such phrases are lacking, a piece of software should provide statements such as “Remember that you’re a team, so applaud your successes.” as well as words of encouragement to prompt celebration and verbal agreement.

Another option is to ensure players have opportunities to perform praise-inspiring feats within the software, prompting applause from peers.

3.2.2 Coordination

The next component in our taxonomy is *coordination* (Figure 3.4): the methods that groups use to organize group members and their actions. Without coordination, a team cannot effectively distribute sub-tasks to maximize group performance, nor is it trivial to respond to environmental changes.

One subcomponent of coordination is the *organization* of group members. Whilst the implied strengths and weaknesses of roles help facilitate quick and optimal task assignment, member roles also invoke an implied authority, which limits inter-member disputes. To remain effectively organized, group members should define [24, 93, 120] and fulfill [93] designated roles, even under duress [77].

To assess organization, we can combine a count of key phrases like "You be the ... while I be the ..." with the completion speed for each role's progress events. If help is needed, software can provide encouragement through statements such as "Remember that you have responsibilities." or an explanation and demonstration of what each role entails. When roles are explicit, it is best to build them into a piece of software, giving each player a different set of possible actions.

After role designation, a group can begin to approach an impending task, taking into account the problem, possible solutions, and group member strengths and weaknesses. This *strategizing* includes how well group members communicate [20] and create strategies [2, 19, 20, 21, 24, 31, 32, 41, 93, 96, 94], in addition to how well they understand [20, 21, 32, 41, 68], select between [2, 31, 32], and execute [19, 20, 21, 41, 68] those strategies. Other aspects of strategizing include decision making [77], prioritization [31, 32, 109], and

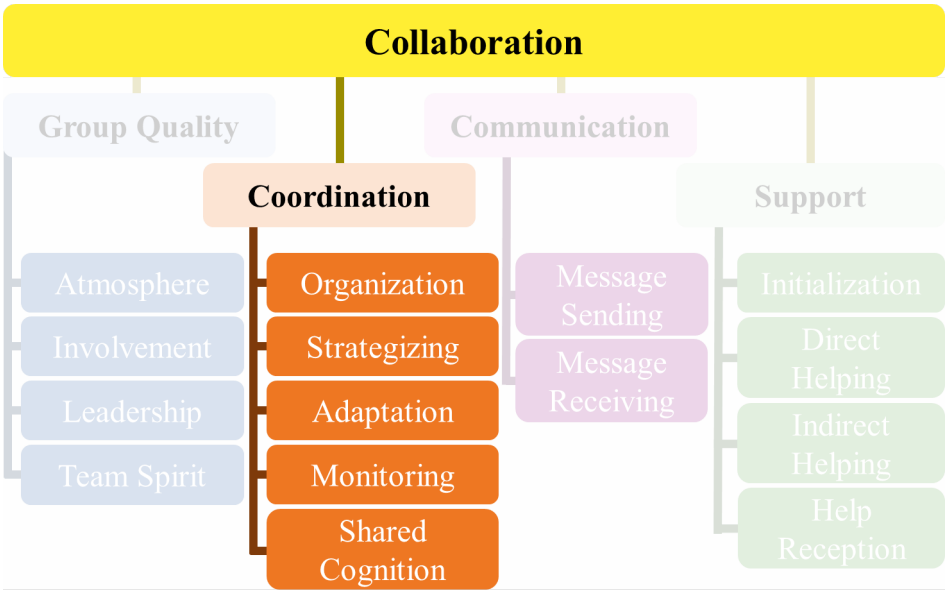


Figure 3.4: Coordination

resource management [2, 31, 70, 96, 109].

By combining topic and speaker segmentation, we can use audio data to determine the percentage of time each player spends discussing strategies. To test strategy selection, we can look for instances of players either succeeding quickly, suggesting an optimal strategy choice, or instances of a player failing several times, discussing strategy with others, and then trying a different approach, showing that a flaw was discovered and an alternative selected. We can also gauge resource management by noting when a player uses a shared resource that someone else needs more. To promote strategizing, software should include challenges that require complex tactics and when necessary may supply scaffolding statements like "If you're having trouble, you may want to come up with a plan."

Often an initial effort is not enough; *adaptation* is needed to re-optimize organization [24], strategies [2, 32, 56, 68, 70, 77, 94, 120], and resource distribution [68] as circumstances change. To adjust accordingly, groups must anticipate [68, 94] or identify [2, 56, 68] the need for change and effectively implement it, possibly by creating an inno-

vative approach [94].

To assess adaptation, we can re-use our approach toward strategizing, except now we focus on events directly after circumstances change within a piece of software. To prompt adaptation, we can stress when such changes occur through visual cues or statements such as "Maybe we need to re-assess our strategy." In addition, software developers should include contextual changes that necessitate strategy modification, ensuring players gain experience in adaptation.

To help identify when adaptation is necessary, groups can use *monitoring*, which involves re-evaluating a strategy [31, 32] and supervising group functions and influences, such as standards compliance [20, 21, 31, 32], boundaries [20, 21], roles [20, 21], environmental changes [94], and adherence to the chosen strategy [20, 21, 41]. Monitoring also includes seeking feedback [67] and keeping track of the big picture [20, 21, 56], as well as overseeing [67, 70, 94] and reflecting on [77] the actions of group members and reviewing past decisions [109] to provide feedback on performance [67, 70, 77, 103, 109].

We can identify monitoring through keyword analysis, looking for reflective phrases like "You were...", "You didn't...", and "That's because you..." Software should support monitoring through additional information in the user interface and through textual prompts that invite re-evaluation like "Maybe you should think about why this isn't working."

Through strategizing, adaptation, and general interaction, group members form a shared mental model [41, 103], also known as a joint problem space [92, 107], of the situation [56, 120], task [20, 92, 107], goals [20, 92, 107], available actions [92, 107], and the relations between these notions [92, 107]. This *shared cognition* allows the group to function as a single entity. Through information exchange [2, 31, 109, 94], sometimes via the articulation of expectations [68], group members formulate an understanding of others' opinions [24], values [24], aptitudes [24], and attitudes toward one's role in the group [24]. Monitoring shared cognition involves ensuring no discrepancy between individual mental models and

making adjustments if necessary [56, 107].

Because a shared mental model is intangible, we must use audio analysis to infer understood meaning by looking for clarification phrases and prompts such as "I thought you meant...", "What I mean is...", and "What do you think?" If such expressions are not forthcoming, we can provoke deeper thought through statements like "Make sure you all understand each other."

3.2.3 Communication

Coordination is usually achieved through effective *communication* (Figure 3.5). While this component does not encompass task-specific content, such as strategies, it does include considerations before message sending, apt response to message reception, and the appropriateness of message content. Without adequate communication skills, group members cannot articulate thoughts, which inhibits the ability to form a shared mental model and prevents the creation of an optimal strategy based on the informed opinions of group members.

Sending a quality message is non-trivial. Messages should be original [4], precise [20, 21], accurate [77], complete [77], timely [20, 21, 77], and with information in the correct order [70]. A message sender should consider conformity to the current discussion [4], terminology [67, 70], content sensitivity [32], aptness of dissemination [77], distribution method [67], recipients [77], and the differences in delivering it verbally (with loud and distinct speech [70]) vs. non-verbally [32].

To determine the quality of a message, we should focus on the responses of message recipients. If the next set of statements have a confused or questioning tone, it suggests that the message was unclear. To remedy such behavior, we can present prompts such as "Before you speak, double check that what you plan to say is clear and to the point."

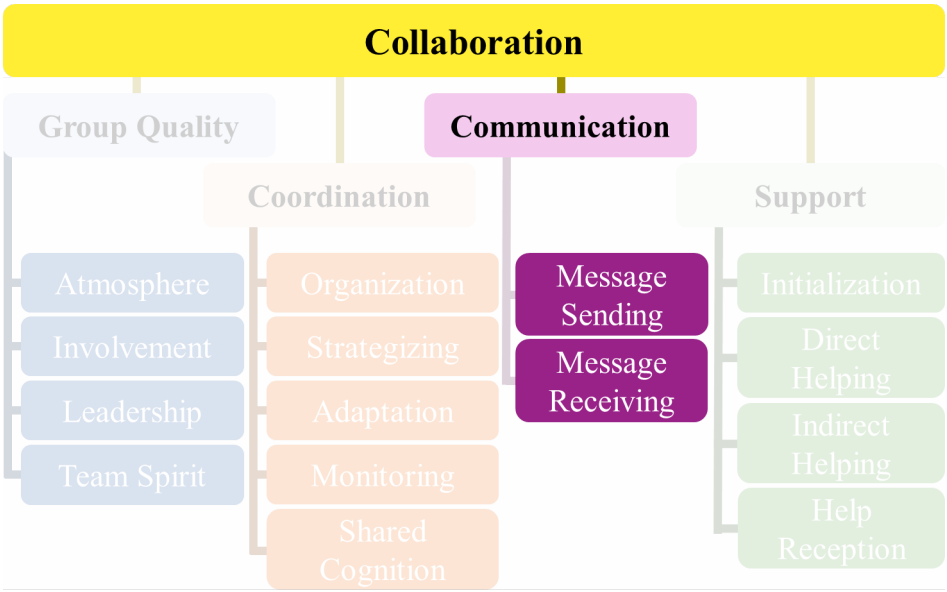


Figure 3.5: Communication

After a message is sent, the sender should confirm it reaches its destination [2]. In return, when *receiving* a message, the recipient should pay attention [24], confirm receipt [67], verify no information was lost in transfer [70], and ask for clarification [67, 70, 77, 103], justification [103], or elaboration [103] if necessary. A discussion concerning the contents of the message may ensue afterward [77].

We can identify a recipient's low attention level through instances where a message is spoken loudly but the recipient either does not respond or replies with a sound or phrase indicating the need for repetition, such as "Huh?" or "Did you say something?" Evidence of receipt confirmation includes statements like "Thank you." or sentences that re-use terms in the message, suggesting a response. To promote reception etiquette, we can scaffold with phrases like "It may help to listen to what everyone has to say." It also helps to make the challenges in a piece of software thought-provoking.

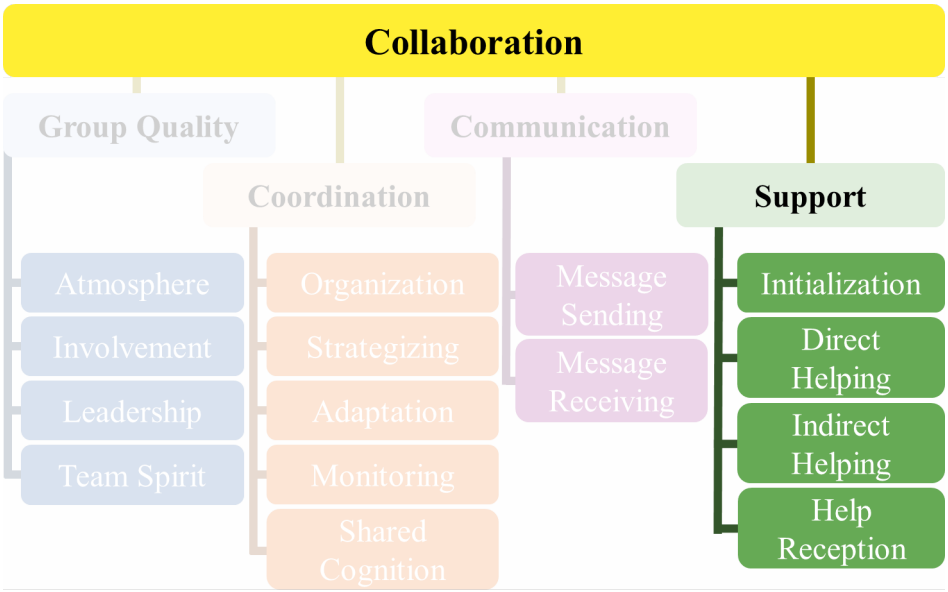


Figure 3.6: Support

3.2.4 Support

The final component of collaboration is *support* (Figure 3.6): group members helping each other complete a task [21, 24, 31, 32, 34, 67, 68, 70, 80, 84, 94], satisfy a role [68, 94], take on a responsibility [68], or understand a concept [103]. The quality of support is based on many factors, including an individual's willingness [41, 78, 79], time [94], resources [94], and ability [41, 67, 94] to help, as well as a person's understanding of the tasks assigned to groupmates [41]. Without support, group members cannot build off of each other's experiences and knowledge, meaning time is wasted learning without a teacher or mentor and re-developing solutions for obstacles overcome in the past.

Help must first undergo an *initialization* during which a group member recognizes [84] and offers help [6, 32, 78, 96] or someone asks for help [34, 41, 70, 70, 96, 94] and a peer agrees to provide it [6, 20, 78, 84, 96]. In either case, it is important to retain a positive atmosphere, otherwise derogative comments and condescending reprimands discourage help requests.

To identify when help is initialized, we can focus on inquiry phrases in audio data, such as "Could you help me?" and "How am I supposed to do this?" We can then look for positive responses like "Sure!" or an agreeing head nod in the video data. A pause in computer input suggests that the request is genuine, as the requesting player does not know what to try next and the helping player has just shifted attention away from a task. To promote help requests, we can present statements like "If you're stuck, it may help to ask your partner." We should also ensure that software provides challenges difficult enough to warrant such requests.

After initialization, the help process itself can commence. There are two types of helping: direct and indirect. *Direct helping* involves explicit aid through the supply of instructions [6, 78, 94], an explanation of what to do next [70], or the partial [6, 84] or total [6, 70, 84, 94] completion of a task assigned to another group member. Unfortunately, with direct helping, the person being helped forgoes the knowledge and experience often gained from personally overcoming a challenge.

A better alternative may be *indirect helping* through resource sharing [6, 34, 78, 79, 84, 94] or by providing guidance [67, 68, 94], education [109], structure [67], suggestions [70, 94], advice [94] or constructive feedback [94]. This type of helping is especially advantageous in error correction, when a group member can provide guidance on error detection [80] or suggest reassessment to prompt someone to find her own mistake [70, 80]. When pointing out a mistake or potential mistake [94], a person should be nice [70] and avoid doing it verbally, which attracts more attention than gestures [70].

Aside from audio cues such as "Let me show you." and "What do you understand so far?", we can assess helping through the identification of assistance patterns. Sometimes a player repeatedly fails at an action and then stops as a group member talks or demonstrates the action before the player tries again, this time in a different manner. The demonstration itself is often characterized by slow speech and methodical software interactions. As with

help initialization, we can promote direct and indirect helping by coupling challenges that necessitate aide with scaffolding statements like “Helping others is a great way to make sure you understand the material.”

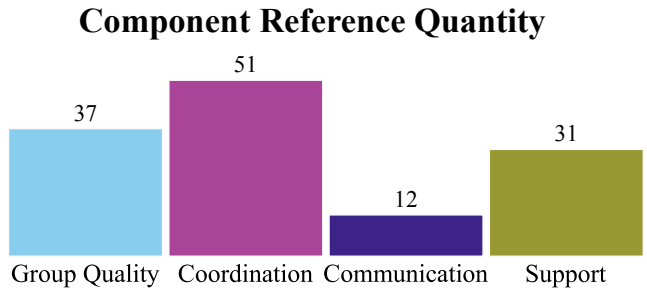
During *help reception*, a group member should not be afraid to ask if an error occurred [70], should try to understand why something is wrong [70], and should thank anyone that helps [70, 80]. Attention and courtesy are ways to show appreciation and respect, which contributes to a more positive atmosphere.

We can evaluate help reception with respect to two sources of aid: other players and scaffolding via software. With other players, we should look for key phrases like “Thanks.” and “Is this what you mean?” If necessary, we can provide scaffolding prompts like “Asking for clarification is better than not understanding the concept.”

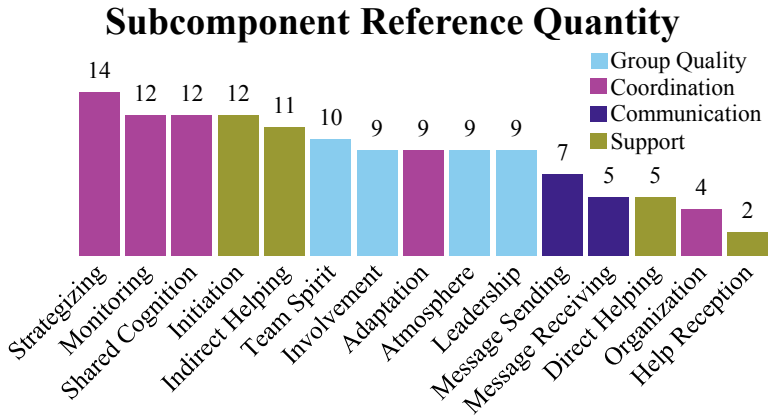
With in-game scaffolding, we should focus on responses within the software. Does a player dismiss the scaffolding immediately, ignoring any help it may provide? Does the player change behavior in the software, such as approaching a problem differently? We can also look for verbal responses like inter-player discussion following the prompt. If a player consistently dismisses scaffolding, we may need to expand the scaffolding system to include more modalities, such as audio, pictorial, and textual.

3.3 Analysis

To inform the collaboration research space, its strengths, and its deficiencies, we chose to explore the patterns and outliers in our taxonomy and its source literature (Figure 3.7). Papers included in the literature search were based on our discovery of seminal papers in the collaboration research space, with a focus on those relevant to our research interest in collaborative scaffolding and play. A naïve comparison of components and subcomponents is meaningless because the breadth of each component and subcomponent varies, so we



(a) Component reference count



(b) Subcomponent reference count

Figure 3.7: Number of references

took a hybrid quantitative/qualitative approach.

3.3.1 Components

Of collaboration's four main components, coordination is the most overrepresented, and not just because one of its subcomponents is highly referenced. This attention is unsurprising given that groups are often formed to coordinate the completion of a task. Strategizing facilitates effective task parallelization and responsibility distribution, while shared cognition allows individuals to assign tasks and resources based on known group member merits. The popularity of monitoring, although warranted in some contexts, is somewhat unexpected given its impracticality for projects with short timeframes, which have fewer

opportunities for assessment and behavior modification.

Unlike coordination, communication and its subcomponents are underrepresented. Without adequate communication skills, group members cannot articulate thoughts, opinions, and directions, which makes higher level activities like strategizing and direct helping difficult to execute. However, this universal application means research into communication spans many domains. Although lacking in the collaboration research space, it may be abundant in other research areas.

3.3.2 Subcomponents

One of the largest disparities among closely related subcomponents occurs between direct and indirect helping. Both serve a similar function in the collaborative process, yet indirect helping is referenced more than twice as often. This difference may in part be due to the over-specificity of direct helping. While indirect helping comes in various forms like teaching, advice, and feedback, direct helping only includes explicit instructions or the takeover of a task. We could have merged the two concepts into a single subcomponent to avoid this imbalance, but then we would have been unable to highlight benefits and drawbacks.

Referenced even less than direct helping is group organization, whose underrepresentation is somewhat expected given its limited applicability. Roles are tied to specializations, so casual groups whose members lack diverse strengths and weaknesses, such as those with children, gain little benefit from designation. Furthermore, members of industry and the military often have established functions apart from any single project or task, so role selection does not occur during the collaboration process.

The least referenced subcomponent is help reception, probably in part due to a combination of its specificity and its dependence on existing social norms independent of col-

laboration. Although underrepresented in our particular literature review, a great deal of relevant research exists in other domains. Thus, further inquiry is not as crucial as the numbers may suggest.

3.4 Use in Technology Evaluation

Our taxonomy can serve as an important tool for computer scientists in the fields of Computer Supported Collaborative Learning (CSCL) and Computer Supported Cooperative Work (CSCW). Because of the dissimilarities between these disciplines, collaboration plays a different role in each. To maximize the utility of our taxonomy, it helps to understand these differences and their effects on evaluation.

As Koschmann defined it, CSCL "is a field of study centrally concerned with meaning and the practices of meaning-making in the context of joint activity and the ways in which these practices are mediated through designed artifacts" [58]. According to Suthers [105], an important focus for this area is the interactional accomplishment of and technology affordances for intersubject learning. Because the goal of technology in CSCL is to facilitate edification, context and educational content plays a large role and needs to be balanced with collaborative skill support. Within this field, our taxonomy can help explore how components of collaboration affect elements like educational scaffolding and each student's zone of proximal development.

In contrast with CSCL, CSCW is "an endeavor to understand the nature and requirements of cooperative work with the objective of designing computer-based technologies for cooperative work arrangements" [100]. Research in this domain mainly focuses on adults in a corporate setting. Because the goal is task completion, rather than learning, content and context are overshadowed by the need to support collaborative functions that improve workflow. Researchers should use our taxonomy to ensure that people are fulfilling neces-

sary roles by focusing on things like group dynamics and collaborative functions.

3.5 Implications

Because of its diverse source material, our taxonomy is applicable to many research areas and subject demographics. Its multiple levels allow for varied scope, while its hierarchical structure lends to a deeper understanding of subcomponent relations. In addition, it can be combined with known methods of evaluation to explore topics such as:

- Strategy creation and retention, measuring quality, complexity, and appropriateness.
- Participation improvements, measuring message sending, team spirit, leadership, and number of utterances.
- Communication improvements, perhaps even during non-group activities, measuring communication, voice volume, and how often an individual answers questions.
- Support improvements, measuring support, team spirit, and individualism-collectivism.
- Organizational improvements, measuring coordination and leadership.
- Task-based improvements, measuring strategies and contribution.

3.6 Research Direction

For this thesis, we chose to focus on support improvements, and in particular on instances of performing part of a task for a group member (direct helping) and providing guidance to a group member (indirect helping). We left out help initiation and reception because they are more closely tied to group atmosphere, which is outside the scope of our current research interests.

4 Study Development

In this chapter, we outline the randomized mixed-factorial study we developed and performed on-site with local middle school students. The study addressed the following research questions, all of which are based on the direct and indirect helping subcomponents of our collaboration taxonomy:

Question 1: Does collaborative scaffolding in an educational game promote more instances of performing part of a task for a group member (direct helping) than non-collaborative neutral scaffolding in the context of the game?

Question 2: Does collaborative scaffolding in an educational game promote more instances of providing guidance to a group member (indirect helping) than non-collaborative neutral scaffolding in the context of the game?

Question 3: Does collaborative scaffolding in an educational game promote more instances of performing part of a task for a group member (direct helping) than non-collaborative neutral scaffolding in the context of a collaborative classroom activity?

Question 4: Does collaborative scaffolding in an educational game promote more instances of providing guidance to a group member (indirect helping) than non-collaborative neutral scaffolding in the context of a collaborative classroom activity?

We used questionnaires, audio recording, keylogging, and direct observation to gather information about helping behaviors before, during, and after students played an educational video game with collaborative scaffolding [61]. As part of the process, we developed Likert scales, modified an existing video game, and recruited students at a middle school and an afterschool program.

4.1 Design

We used a randomized mixed-factorial design with one control group and one treatment group. Students played in randomized pairs with varied gender matchups. Both control and treatment groups played the same educational video game but each encountered a different type of scaffolding.

We can think of scaffolding as being on a scale from positive to negative. Positive collaborative scaffolding benefits students, reflecting the traditional definition of the term. It includes statements such as "Remember each of your abilities, and use them to help each other." and "Sometimes things are easier to do together." On the other end of the spectrum, negative collaborative scaffolding hinders students with statements such as "You don't need anyone's help." and "Working with others is a waste of time." Between the two there is neutral scaffolding, which neither helps nor hurts students. Neutral scaffolding often takes the form of irrelevant and non-controversial phrases including "It must have taken a long time to build this place." and "It sure is bright outside." Our treatment group encountered context-specific positive collaborative scaffolding as outlined in Appendix E, Table E.1. The control group saw neutral scaffolding statements based on the same contextual triggers, but with content drawn sequentially from a list of phrases (Appendix E, Table E.2). By presenting scaffolding to both groups, we aimed to account for its disruption to gameplay.

During each trial, students started by taking a short pretest that covered their collabora-

tive experiences and attitudes toward collaboration, as well as their knowledge of our video game's science topics. They then split into small groups and performed a simple collaborative classroom activity during which we took note of direct and indirect helping via audio recorders and several observers. After the activity, we split students into a control and a treatment group. We randomly assigned pairs in each to play the appropriate version of the game before having students individually fill out a short questionnaire on engagement and motivation. Keeping students in the control and treatment separate, we formed small groups to once again participate in the collaborative activity. Students finished by filling out a posttest pertaining to collaboration and one on knowledge of the subject area.

4.2 Questionnaires

4.2.1 Attitudes Toward Collaboration Pretest and Posttest

To measure subject attitudes toward collaboration before and after play, we designed a pretest (Appendix A) and posttest (Appendix D) with 7-point Likert scales. Our items were based on Johnson and Norem-Hebeisen's [50] *cooperative interdependence* scale, which has been tested for reliability (22 items, Cronbach's α = 0.84). We modified the *liking to cooperate* subscale to develop our *liking to collaborate* scale. "Students" became "people" to extend applicability, and we split compound items to simplify the questionnaire, producing the following scale:

- I like to help others learn.
- I like to share my ideas with others.
- I like to share my materials with others.
- I like to cooperate with others.

We also created a *valuing collaborative learning* scale based on the *cooperative interdependence* scale's *valuing cooperative learning* subscale. As with *liking to collaborate*, we converted "students" to "people" and divided items. For additional clarity, we also added "I believe" to statements that asked for speculation on the perspective of others. To keep our questionnaire short, we removed "Students learn lots of important things from each other." because of its similarity to "I can learn important things from other students.", resulting in the following items:

- I believe that I can learn important things from others.
- I try to share my ideas with others when I think it will help them.
- I try to share my materials with others when I think it will help them.
- I believe that people should help each other learn.

Prosocial tendencies is our last scale and measures propensity toward helping others, augmenting our observations before, during, and after play. Two items were based on the statements "I tend to help people who are in a real crisis or need." and "I never hesitate to help others when they ask for it." from Carlo and Randall's [18] *Prosocial Tendencies Measure*. Although the authors did not report the overall reliability of their measure, the reliability of its six subscales is as follows: public (4 items, Cronbach's α= 0.78), anonymous (5 items, Cronbach's α= 0.85), dire (3 items, Cronbach's α= 0.63), emotional (4 items, Cronbach's α= 0.75), compliant (2 items, Cronbach's α= 0.80), and altruism (5 items, Cronbach's α= 0.74). The rest of the items, as listed below, came from our other scales:

- I like to help others learn.
- I try to share my ideas with others when I think it will help them.

- I try to share my materials with others when I think it will help them.
- I believe that people should help each other learn.
- I try to help people who need it.
- I don't hesitate to help others when they ask for it.

We evenly divided a mutually exclusive subset of items from all scales across our pretest and posttest.

4.2.2 Knowledge Pretest and Posttest

For the knowledge pretest and posttest, we used the same set of questions (Appendix B) based on the content of the educational game used in our study. We included multiple choice and matching questions, but avoided short answer and essay questions to minimize fatigue and standardize grading. Because we did not review answers or directly teach the science topic being tested, improvements when comparing answers pre- and post-play could be attributed to the collaborative video game. The knowledge test was administered at the same time as the attitudes questionnaire, and participants were given seven minutes to fill out both.

4.2.3 Motivation and Engagement

To gauge the impact of scaffolding on motivation and engagement, participants were given five minutes to fill out a 7-point scale Likert questionnaire (Appendix C) immediately after playing the video game. In designing the Likert items, we borrowed heavily from Brockmyer et al.'s Game Engagement Questionnaire [15] (19 items, Cronbach's $\alpha = 0.85$), changing tenses and making minor modifications for clarity to produce:

- I lost track of time.
- I was tired from playing, but only noticed after I was told to stop.
- I felt like I was actually in the game world.
- My interactions in the game seemed real.
- I cared about the outcome of the game.
- When someone talked to me, I didn't notice because I was focused on the game.
- I lost track of my real surroundings.

We also modified several items from the Intrinsic Motivation Inventory's [97] interest/enjoyment and effort/importance dimensions to develop the following:

- I put a lot of effort into playing the game.
- I tried very hard to win the video game.
- I enjoyed playing the game.

A subset of these IMI dimensions with similar minor modifications was found to be fairly reliable (overall: 18 items, Cronbach's $\alpha = 0.85$; interest/enjoyment: 5 items, Cronbach's $\alpha = 0.78$; effort/importance: 4 items, Cronbach's $\alpha = 0.84$) [66].

From all of these items, we created the following Likert scales:

- *Immersion / Presence*: How "in the game" the participant felt
  - I lost track of time.
  - I felt like I was actually in the game world.
  - My interactions in the game seemed real.

  - I lost track of my real surroundings.

- *Effort / Importance*: How much the participant became invested in the game
  - I started caring about the outcome of the game.
  - I enjoyed playing the game.
  - I put a lot of effort into playing the game.
  - I tried very hard to win the game.

- *Engagement*: How real in-game interactions became during play
  - My interactions in the game seemed real.
  - I lost track of my real surroundings.
  - I started caring about the outcome of the game.
  - I put a lot of effort into playing the game.
  - I tried very hard to win the game.

- *Flow*: How involved the participant became during play
  - I lost track of time.
  - When someone talked to me, I didn't notice because I was focused on the game.
  - I was tired from playing, but only noticed after I was told to stop.
  - I lost track of my real surroundings.

4.3 Classroom Activity

To gather data on student interactions before and after play, we developed our own collaborative classroom activity. The rules had to be simple enough to explain and understand

within minutes to ensure our study stayed within an hour timeframe, and all group members had to have their own tasks to encourage participation. The activity also had to have some level of difficulty to impede fast completion and incentivize student interactions. Because we were interested in exploring willingness to collaborate, we furthermore needed to guarantee an opportunity for, but no requirement of, collaboration.

We decided on a Scrabble-like activity. Each group member received a set of letter tiles of a certain color to create his or her own crossword puzzle containing science-related terms. Optionally, each member's words could interlock with those of other group members, but only in one place per word. The activity was considered complete when no one had letters left.

Our design followed several of the collaborative game patterns and lessons outlined in the Background chapter. For example, the tiles served as a limited resource [102] that was different for each student [117] and that others could not control [117]. Moreover, connecting words was mutually beneficial [89], while color coordination allowed students to trace payoffs to decisions [117].

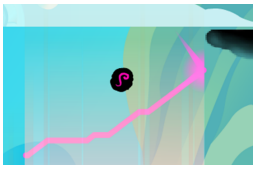
4.4 Video Game

For the video game itself, we modified the Singapore-MIT GAMBIT Game Lab's *Waker* (`http://gambit.mit.edu/loadgame/waker.php`), a 2009 action-puzzle Flash game that teaches displacement and velocity. We chose this game because of its light-weight nature, educational content, and potential for rapid modification. Since it is a Flash game, we were able to easily embed it in a webpage, which could be accessed from all school computers without having to worry about installation restrictions. To adapt the software to our needs, we made the following changes:

- *Multiplayer* - The game was single player, so we implemented a second avatar. As



(a) An avatar picks up an orb of its color.



(b) A graph in a predesignated area is slowly drawn based on the the avatar's movement.



(c) When the avatar drops its orb, the graph solidifies to form a path that everyone can walk on.

Figure 4.1: *Waker* core game mechanic.

Figure 4.1 depicts, the core mechanic was ideal for multiplayer play. We added color coding to differentiate player avatars and the objects each could pick up.

- *Levels* - To complement the multiplayer nature of the game, we created new levels based on the originals, maintaining similar challenges and difficulty levels.

- *Demo* - To help players grasp the game's mechanics, we designed and implemented an informative animation.

- *Scaffolding* - The game had a hint system in place, but it was not robust enough for our purposes. We developed our own system including:

  - The use of in-game events as triggers
  - The creation of invisible zones to be used as triggers
  - Trigger interrupt events, so scaffolding can be canceled if players no longer need it by the time it is scheduled to appear
  - Visualization in the form of thought bubbles with textual scaffolding
  - Visualization in the form of avatar ghosts to guide players

- *Logging* - For data collection, we constructed a logging system that tracks in-game events and player inputs.

4.4.1 Scaffolding

Design

As our third major contribution, we developed an infrastructure for collaborative scaffolding cues (Figure 4.2). As the figure depicts, scaffolding is trigger by an event, which can be one of three types. A spatial event occurs when an avatar enters or leaves a section of a level as defined by a bounding box, while a game event occurs when an interaction within the level occurs, such as an avatar picking up an orb. The third type of event is related to low level input from mouse and keyboard. Events can have associated players, values, and objects in the level. For example, a particular player may pick up a particular orb. Conversely, the start of the level is player independent, but has a level number to designate its position in game progression.

Once a trigger event occurs, a period of delay begins. The length of the delay is based on either a chosen temporal length or a count of trigger event repetitions. During this time, scaffolding can be interrupted, preventing the visualization from occurring. The interrupt event has the same types of properties as the trigger event, although the properties themselves do not have to be the same.

If no interrupt occurs, after the delay a scaffolding visualization appears. The visualization comes in two forms: text and animations. Textual scaffolding appears as a thought bubble originating from one of the avatars (Figure 4.3a), while visual animations appear as color-coded ghosts representing either (or both) avatars (Figure 4.3b). Because of the usefulness and shorter creation time of textual scaffolding content, visual animations were a low priority for us. Although we chose to omit them from our current trials, we plan to

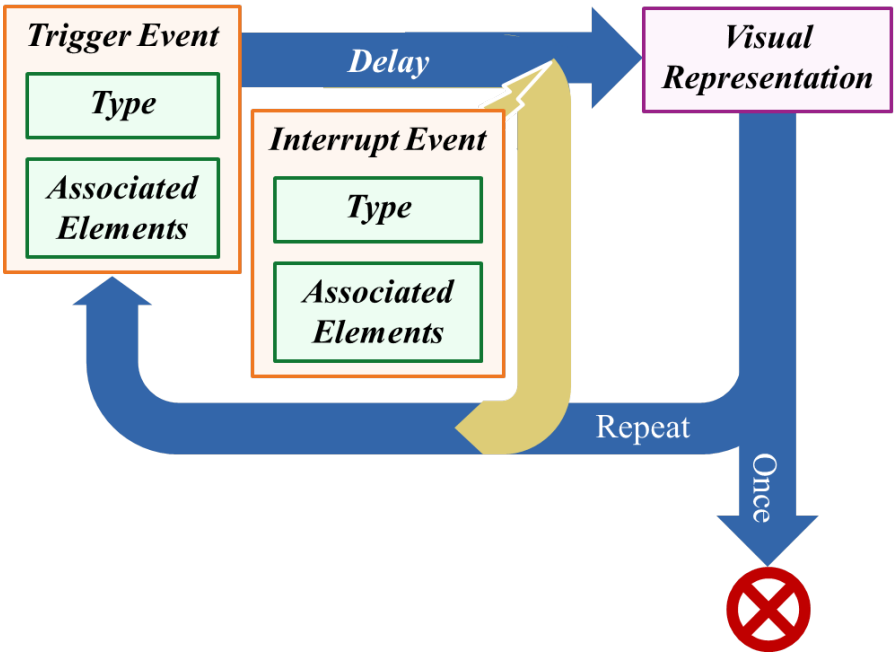
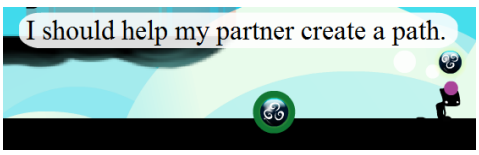


Figure 4.2: Scaffolding infrastructure



(a) Textual



(b) Animation

Figure 4.3: Types of scaffolding

incorporate them into future studies.

After the visual scaffolding appears, there is an option to disable the scaffolding cue so it does not trigger again in the future. In our own system, we often chose this route because we did not want players to get frustrated with seeing the same scaffolding over and over in the short period of time they had to play.

Challenges

During the design of our scaffolding instrumentation, we ran into many challenges. The first were in the content of the textual scaffolding, which couldn't be too repetitive and had

to be direct enough to make an impact without being overly direct like an order. Timing also played a role, as we had to refine appearance rate and equalize that rate among players. The latter was nontrivial because the probability of a scaffolding cue being triggered varied across different events. The visual appearance also had to be carefully considered. We didn't want to distract players too much in the middle of an action, but we wanted to grasp player attention to avoid inattentional blindness [63]. We opted to display a series of small white bubbles next to the target avatar, which the player was probably focused on, with a larger thought bubble at the other end. We also didn't want to cover important visual information about the level, which is why we made our scaffolding visualization translucent.

Once the design of our system was complete, we still had technological challenges to address. Most stemmed from the creation of new features, such as spoofing input for the demo and visual scaffolding, as well as detecting the entrance and exit of a particular area, which was a necessity for the capture of spatial trigger and interrupt events. Additional challenges included the dynamic positioning of textual scaffolding thought bubbles and the design of the scaffolding function used to add scaffolding cues to a level, which had to address and have parameters for the trigger event, interrupt event, timing, scaffolding appearance, and whether the scaffolding could appear again.

4.4.2 Logging

Design

Our forth contribution is a data collection infrastructure. As participants played via web browser, our system stored keyboard events, mouse events, mouse position, and player position. We also logged several types of in-game events (Figure 4.4) to monitor and extrapolate data like how far players progressed, how much players participated in playing,

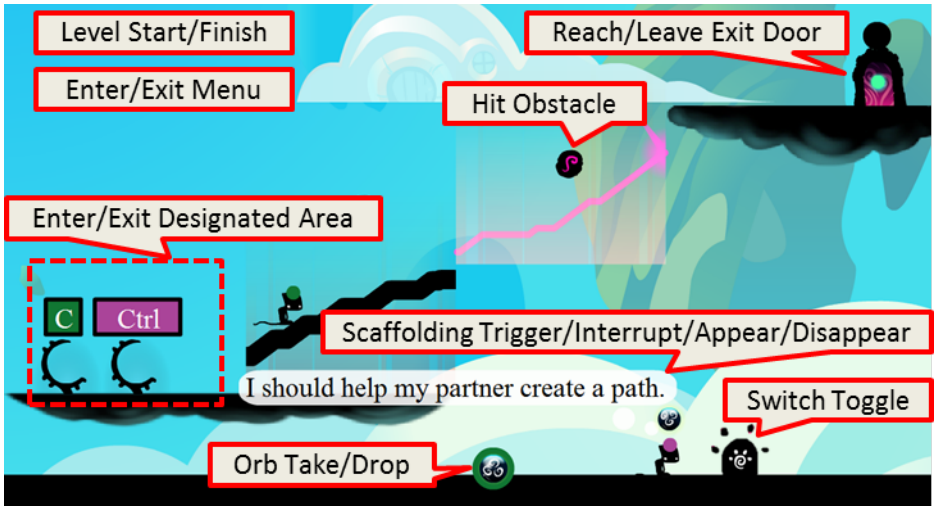


Figure 4.4: In-game events

and if players payed attention to the scaffolding.

Challenges

Designing the scaffolding required careful consideration and refinement. We needed to standardize names and information formats for all event types to support code modularity and accurate event capture across our code base. In addition, we had to consider the meaning of events for future data analysis. Because too much collection caused visual lag, we weighed the relevance and minimum useful frequency of data against performance impact.

We also encountered challenges on the technical side. Some events already existed but were not set up for capture, while others we implemented ourselves. Furthermore, they were all scattered throughout the software, requiring a search for their location, as well as the creation of event hooks. Determining a storage methodology ended up being one of our biggest technical challenges. We had several alternatives, each with their own drawbacks:

- *Adobe AIR*: Flash projects can be converted into Adobe AIR projects, which allow local data storage. However, they also require installation onto every machine, which

can be time consuming and potentially impossible given the strict permissions school computers tend to have.

- *Local Files*: Data can be stored in local text files without conversion to Adobe AIR, but it requires distracting user prompts. If these prompts come up during play, they can lead to annoyance, confusion, and user-induced issues.

- *Local Storage Objects (LSOs)*: Also known as *Flash cookies*, LSOs can be stored and updated locally without user prompts, but by default they are limited in size to 100KB. If the limit is not increased ahead of time on every machine, a confusing prompt comes up during play requesting more space.

- *Server*: Flash projects can be embedded in a webpage, facilitating access from any browser and allowing data to be stored in a SQL database on a webserver. Because the project does not run locally, there is no need for instillation. However, not all schools have reliable internet access, which is required to continuously send information between the local machine and the server.

- *JavaScript*: To avoid the need for internet access, Flash projects can be embedded into a webpage and then transferred to school machines via physical media. JavaScript code can log data directly onto the webpage, although the information would later have to be copied manually into permanent files. This methodology is prone to error, since closing the tab or browser results in complete data loss.

We initially decided on LSOs because installation is not necessary, there is no reliance on internet access, and preparation can be done beforehand to avoid the need for player input. However, when checking the technology on-site at our first trial, we discovered that Flash was disabled on all machines. Because the study location had reliable internet access, we chose to instead store the game on a server and embed it in a webpage.

4.5 Logistics

4.5.1 Challenges

Our first logistical challenge was privacy. Because we ran trials with children, it was important to balance data and confidentiality. Aside from the usual precautions of avoiding names and anonymizing data before analysis, we also chose to limit our data collection to computer logging, audio recording, and direct observation to avoid storing video.

With no pre-existing affiliations, recruitment was also non-trivial. We needed to be on site for the trials, so schools and programs had to be within driving distance. In addition, to keep the demographic as neutral as possible we had to avoid exceptionally low and high income areas, further limiting our options. To aid in identifying potential locations, we developed a database of schools and programs based on information from the National Center for Education Statistics and the New York State Education Department. From the database we selected applicable schools and sent out recruitment emails to science teachers. Although the affirmative response rate was low, our search was successful.

After recruitment, scheduling became our main concern. We could not run trials during holidays, standardized testing days, or exams, and often class length and meeting times were too short to accommodate our study. For one of our testing sites, we had to split trials across two consecutive days, losing data from an absence on the second day.

4.5.2 Locations

We ran our first set of trials at Midlakes Middle School (Table 4.1), a public magnet school in the Phelps-Clifton Springs Central School District in Ontario County. As of the 2013-14 school year, it has a student/teacher ratio of 14.57 and serves 277 seventh and eighth graders, about 36% of which are eligible for free or reduced lunch [72]. Our trials spanned

Table 4.1: Demographics

Location	Trial	N	Gender		Age		
			Male	Female	Min	Med	Max
Midlakes Middle School	1	15	6	9	12	12	13
	2	8	4	4	13	13	14
	3	15	8	7	13	14	14
	4	4	2	2	12	12	14
	5	10	6	4	12	12	13
Kids on Campus	1	10	9	1	10	11	12
	2	9	7	2	10	11	14
Totals		71	42	29	11	13	14

five classes, each lasting two consecutive days due to class length.

We ran two more trials at the Rochester Institute of Technology's Kids on Campus program (Table 4.1), a month long program during which students from local elementary, middle, and high schools enroll in all day STEM classes. Although the program costs money, scholarships are awarded based on need, meaning our subject pool was not overly biased toward the wealthy.

5 Results

As our fifth major contribution, we gathered and interpreted data on local middle school students. Below we outline the results, augmenting them with a discussion in the next chapter. Although there were 71 participants across all experiments, we had to eliminate some of the data we collected for various reasons: one person was absent on the second day, several participants chose not to fill out certain posttests, one group had three people playing the game simultaneously due to an odd class size, and for some people there were technical difficulties during play. One student withdrew consent, so we removed them even from our total N discussed in the Study chapter.

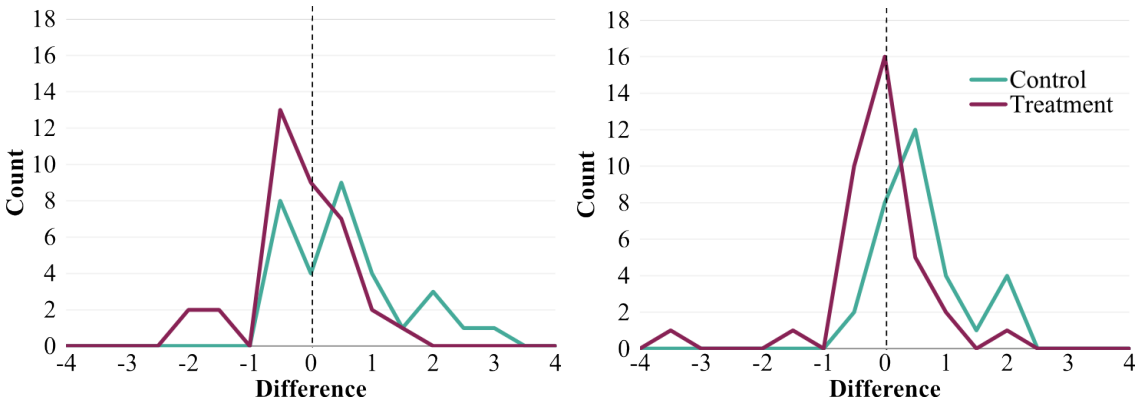
5.1 Questionnaire and In-Game Data

We used t-tests and means-based calculations to compare the questionnaire data of our control and treatment groups (Table 5.1). Collaboration and motivation questionnaires had scales based on 7-point Likert items (Appendix A, Appendix D, and Appendix C), while the knowledge pretest and posttest was graded as a percentage out of 100 (Appendix B). Students in both groups began with similar levels of science knowledge and opinions on collaboration. When compared to neutral scaffolding, collaborative scaffolding did not affect the educational impact of the game, but there was a statistically significant difference in influence on *liking to collaborate* and *prosocial tendencies* (Figure 5.1).

Table 5.1: Questionnaire analysis

			Control		Treatment					
Questionnaire	**Timing**	**Scale**	***M***	***SD***	***M***	***SD***	***df***	***t***	***p*** [†]	***d***
Collaboration	Pre	Liking to collaborate	5.45	1.08	5.64	1.039	65	0.72	0.47	0.18
		Valuing collaborative learning	5.81	1.030	5.75	0.87	65	0.24	0.81	0.059
		Prosocial tendencies	5.57	1.023	5.75	0.83	65	0.80	0.43	0.19
	Pre/Post Difference	Liking to collaborate	0.52	1.029	−0.25	0.82	65	3.40	0.001**	0.83
		Valuing collaborative learning	0.27	1.079	−0.069	0.82	65	1.48	0.14	0.36
		Prosocial tendencies	0.41	0.74	−0.29	0.86	65	3.54	0.0007***	0.87
Knowledge	Pre	Overall	0.63	0.22	0.61	0.23	67	0.37	0.71	0.14
	Pre/Post Difference	Overall	0.011	0.16	0.024	0.13	67	0.38	0.70	0.030
Motivation	Post	Flow	4.26	0.78	3.69	0.96	68	2.70	0.009**	0.64
		Immersion/Presence	5.13	1.14	4.29	1.17	68	3.022	0.004**	0.72
		Effort/Importance	6.015	1.015	5.68	1.045	68	1.35	0.18	0.32
		Engagement	5.63	0.98	4.99	0.98	68	2.70	0.009**	0.65

[†] * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$



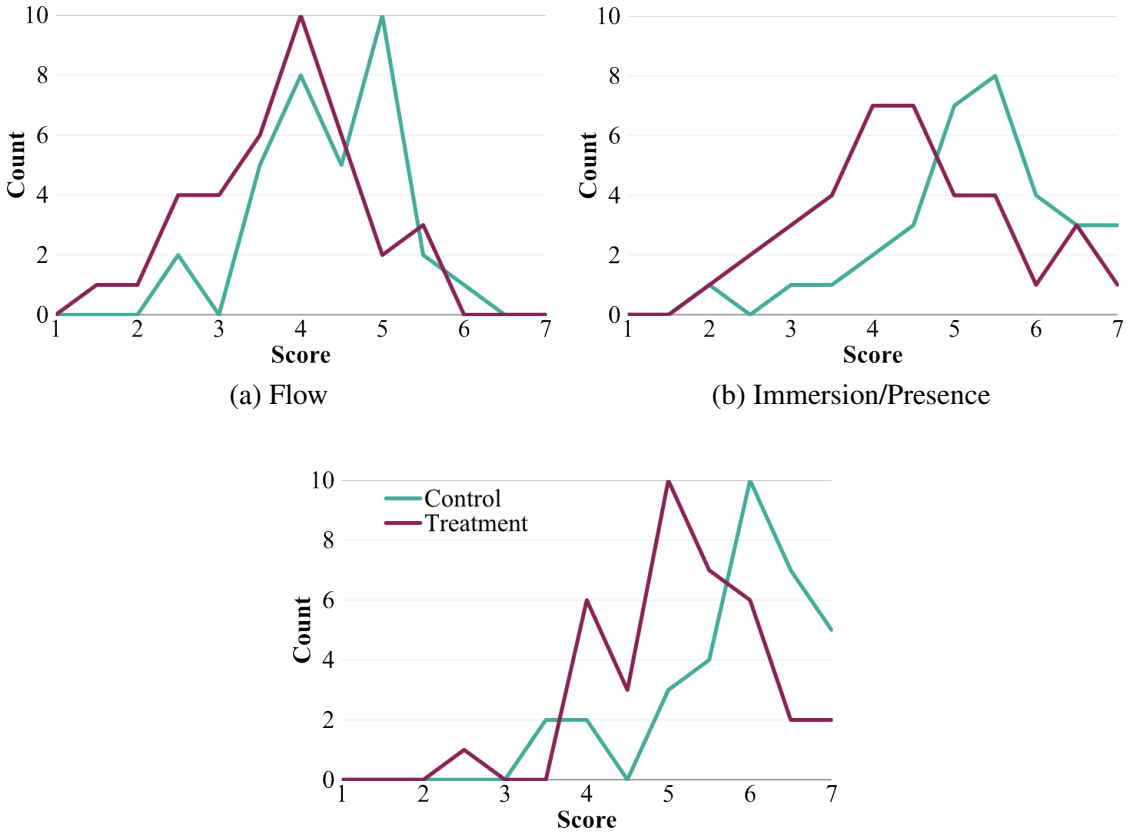
(a) Liking to collaborate (b) Prosocial tendencies

Figure 5.1: Frequencies of Likert scale changes in the collaboration questionnaire.

Paired t-tests revealed that on average, the control group's *liking to collaborate* score increased by 0.52, while the treatment group's *decreased* by 0.25 ($t(65) = 3.40, p = 0.001$). Similarly, the control group's *prosocial tendencies* increased by 0.41, while the treatment group's *decreased* by 0.29 ($t(65) = 3.54, p = 0.0007$). Collaborative scaffolding also had a statistically significantly more negative effect on flow ($t(68) = 2.70, p = 0.009$), immersion/presence ($t(68) = 3.022, p = 0.004$), and engagement ($t(68) = 2.70, p = 0.009$) (Figure 5.2).

Analysis of in-game data led to several more notable statistics (Table 5.2). For example, there were no statistically significant differences in the rate at which student avatars were harmed by hazards. In addition, 29% of participants never beat the first level and another 32% did not complete the second. After accounting for variations on play time, there were no statistically significant differences in student progress across groups.

We also found no statistically significant differences in how often or how long scaffolding appeared for individual players or for groups, nor were there statistically significant differences between key press rate both during and outside of scaffolding. However, through a paired t-test we found statistically significant differences concerning the change in input rate when scaffolding appeared (Table 5.3). The input rate during neutral scaffolding in-



(a) Flow

(b) Immersion/Presence

(c) Engagement

Figure 5.2: Frequencies of Likert scale values in the motivation questionnaire.

creased by 0.33 key presses per second while for collaborative scaffolding it decreased by 0.15 key presses per second ($t(29) = 2.45, p = 0.021$). Looking closer within each group, we found that the change was significant for the control ($t(15) = 2.38, p = 0.031$) but not the treatment (Table 5.2).

5.2 Audio Data

Audio data revealed no statistically significant differences concerning frequency of questions or helpful statements (Table 5.4). However, qualitative results provided hints and insights into student perception and attitudes. For example, in both the control and treat-

Table 5.2: In-Game analysis

		Control		Treatment					
Type	**Measure**	***M***	***SD***	***M***	***SD***	***df***	***t***	***p*** †	***d***
Performance	Hazard hits per minute	1.28	1.38	1.40	1.09	61	0.37	0.71	0.094
	Highest level beaten	0.50	0.63	0.47	0.83	29	0.12	0.90	0.045
Scaffolding Appearance	Rate per player per minute	0.19	0.18	0.20	0.16	61	0.25	0.80	0.063
	Rate per group per minute	0.39	0.30	0.40	0.19	29	0.090	0.93	0.032
	% time per player	0.016	0.014	0.016	0.012	61	0.073	0.94	0.018
	% time per group	0.052	0.022	0.053	0.023	29	0.076	0.94	0.027
Key Presses Per Second	No scaffolding	1.82	0.42	1.89	0.40	29	0.46	0.64	0.17
	Scaffolding	2.16	0.61	1.74	0.76	29	1.67	0.11	0.60
	Difference	0.33	0.56	−0.15	0.53	29	2.45	0.021*	0.88

† * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$

Table 5.3: Key presses per second

	No Scaffolding		Scaffolding					
Group	***M***	***SD***	***M***	***SD***	***df***	***t***	***p*** †	***d***
Control	1.82	0.42	2.16	0.61	15	2.38	0.031*	0.59
Treatment	1.89	0.40	1.74	0.76	14	1.08	0.30	0.28

† * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$

ment groups, the helpful statements were almost always in the form of direct help, often through orders with little to no justification.

Some students embraced the game's collaborative nature with statements like "This is so boss. It's like [the collaborative game] Portal." and "This has to be one of the funnest games ever; having like an older kid next to me," while others were not as thrilled. As one student put it: "I wish this wasn't a game of where we have to work together." Overall, students seemed to enjoy the game, as evidenced by laughter and statements like "This is

Table 5.4: Audio analysis

	Control		Treatment					
Measure	***M***	***SD***	***M***	***SD***	***df***	***t***	***p***†	***d***
Questions per minute	0.63	0.38	0.79	0.68	25.30	0.86	0.40	0.29
Helpful statements per minute	5.15	2.60	4.46	1.93	31	0.87	0.39	0.30

† * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$

just fun. I like this game." and "It's a cool game. Can I buy this game ... on [the digital game store] Steam?" Despite the amusement, there was a lot of frustration, partially due to the game's difficulty. Utterances of "This is like the world's hardest game." and "This is impossible. I can't believe we're only on the second level." could be heard, but were sometimes replaced with statements of relief upon success like "I understand now. You're such a tricky little thing." and "Yeah. We're good at this. It took us four minutes, but..."

In the treatment group, students often responded negatively to collaborative scaffolding. It was sometimes read in a mocking tone and responses include statements like "That's stupid." and "Oh, shut up you. You educational game, you." However, occasionally students reacted positively, specifically to prompts encouraging discussion concerning how to get past an obstacle. Statements of "How do we get past these obstacles?" and "OK. Discuss how to get past an obstacle." were heard.

With the neutral scaffolding, there were more responses of laughter and jovial sarcasm, although one student complained "It keeps on saying these edges are slippery!" Such scaffolding also caused confusion among players who interpreted the content as hints. One of the statements we presented was "I hope it doesn't start to rain." to which a student pondered "What's it mean by 'I think it's gonna rain?"'. His partner replied "I think it's gonna rain and you're going to fail." Another group thought the phrase "I sure am fast." meant one player had greater speed abilities than the other.

6 Discussion

Our initial goal was to explore the impact of collaborative scaffolding on direct and indirect helping. We found no significant effects on helpful utterances, and indirect help was almost non-existent in both the control and treatment groups. However, our questionnaire results revealed that an analysis of skill may not be as important or interesting as one of opinion.

Surprisingly, collaborative scaffolding had a negative impact on attitudes toward collaboration, especially when compared to neutral scaffolding. Because both the control and treatment groups played a collaborative game, progressed the same amount, and encountered obstacles at the same rate, this correlation most likely stems from the scaffolding itself and not elements within the game. There are two components to the scaffolding: timing and content. Since there was no difference in how long or often the scaffolding appeared for pairs in either group, the source is most likely the latter.

Students in the control group saw random phrases, but students in the treatment group encountered statements that instructed them to work together. Because our demographic was middle school students and they were paired randomly, the negative impact may have been caused by early adolescent rebellion, which is a natural aspect of child development [17, 76]. Tendencies toward rebellion can stem from influences like family [3] and socioeconomic status [111]. Because we did not collect detailed demographic information, we cannot rule out these external factors.

Another cause could have been reactance. According to the theory of reactance [12], “a

threat to or loss of a freedom motivates the individual to restore that freedom" [13]. That threat is perceived as even worse if there is a noticeable attempt to persuade [13]. Because our scaffolding told students to collaborate and promoted the importance of collaboration, students protected their freedom to choose not to collaborate by doing the opposite of what we suggested. Such responses based on reactance have already been seen with respect to anti-tobacco [69] and anti-drinking [7] campaigns, and reactance may explain low patient adherence to medical treatment [33]. The issue is prevalent enough that minimizing reactance via more subtle persuasion is already being explored in the prosocial game space [53].

Aside from reactance, our participants' proficiency in collaborative skills may have also influenced the potential impact of our scaffolding. We did not measure skills directly, but rather inferred that collaborative skill deficiency prevented progress. However, our audio data revealed that occasionally participants had already discussed solutions or were still in the midst of doing so. The scaffolding only reminded them of their inability to advance without providing any meaningful aid.

It may also be that a more influential factor we did not measure, such as mood and its transfer [5], overshadowed the value of our collaborative scaffolding. For example, some students may have had higher tendencies toward antisocial behavior, which are not necessarily reflected in conscious attitudes toward collaboration. Such propensities are cultivated over time through elements like family, peers, and community [87].

The high difficulty level of our game, as evidenced by the percentage of participants who did not complete the first level, may have also strongly influenced student opinion. Different students have different responses to failure depending on their goals and perceived ability level [28]. Furthermore, the amount of engagement students exhibit when encountering difficulty depends on many factors, such as how the student was socialized over the years and how the student is currently being taught [73]. Because self-assessment

is not always accurate, the results of our engagement questionnaire do not negate this possibility.

Another influential cause may have been group dynamics. Culture influences attitudes and behavior [52], as well as knowledge sharing [9] and the adaptation of new collaboration technologies [95]. Moreover, diversity has been show to affect collaborative behaviors [55]. In our study, utterance rate varied widely across pairs, and interaction patterns spanned not just groups, but entire classrooms. Some groups tended to bounce questions off of one another, in others a single person would constantly instruct a partner, while still others spent their playtime in silence. This variety may also account for the lack of significant differences in helpful utterances, although a small sample size could also be to blame.

Shifting focus to the results of the motivation questionnaire, we can speculate on why collaborative scaffolding had a significantly negative impact on most of the scales. One explanation is that, unlike with the neutral scaffolding, students actually stopped to read the text, resulting in a break in flow, immersion/presence, and engagement. Although not a significant difference, students did provide less input during collaborative scaffolding than when scaffolding was not present. The disruption may be a natural consequence of attracting student attention and a necessary sacrifice for instilling new skills.

7 Future Work

The negative responses to our collaborative scaffolding only bolster the need for further investigation. We plan to make several modifications and improvements to our software and study design. Some encompass research directions we have not explored yet, while others are based on our numerous results.

7.1 Content

One of our most important adjustments will be to update the scaffolding content. Because of the confusion neutral scaffolding caused, we will create all new statements for future studies, filtering potential options based on possible misinterpretations. Our challenge will be in identifying the different types of misunderstandings that can occur and avoiding each accordingly. For the collaborative scaffolding, we will aim to make the content less distracting by experimenting with different fonts and colors. Since we already have the framework for animations, we will also explore alternate modalities.

Other changes to content will reduce frustration stemming from our game's difficulty. To allow participants to enjoy our full set of levels and game mechanics, we will add easier levels to introduce elements more gradually over time. We will also incorporate more direct hints into the collaborative and neutral scaffolding to avoid exasperation from encountering esoteric advice.

7.2 Technology

When implementing our scaffolding content, we found our architecture lacking in robustness. To support complex response timing, we plan to change the definitions of trigger and interrupt objects to reflect sequences of events rather than single events. Moreover, we will include the option of adding Boolean checks, such as if a switch is flipped or if a player's input rate is below a certain threshold. To minimize negative impact on flow, we will also develop a specialized learning algorithm for intelligently fading scaffolding content.

With all of these improvements, our system would still be limited to a single game. To increase the impact of our work and entice others to build off of our research, we will develop a game-independent version of our scaffolding software, complete with a corresponding API for ease of use. The challenge will be in selecting a new platform. Although a stand-alone program that utilizes machine vision is feasible to a limited extent, a more realistic approach will be to create a new component in Unity (`http://unitey3d.com/`), a popular cross-platform game engine used mainly for 3D game development. This component will act as a class that developers can use and modify to suit their needs.

7.3 Study Design

In the future, we plan to modify our design to account for the challenges inherent in our time constraints. During our study, class length was limited and we needed to avoid fatiguing students, so we often ran trials over consecutive days. Many participants did not progress far in the game, and on several occasions we were forced to skip the collaborative activity after play. In upcoming iterations we will mitigate the need to omit procedure steps and allow for breaks to minimize fatigue.

Other limitations arose because our audio and in-game data was not synced. When

technical difficulties occurred, we could use the audio data to identify when they started and when play resumed, but the in-game data could not detect such anomalies. Alternatively, in-game data revealed every time scaffolding appeared, but audio responses to scaffolding could only be identified when participants read statements out loud. For future studies, we will develop a methodology for accurately syncing data modalities, perhaps through supplementary video recordings.

Another of our data limitations was the insufficient amount of demographic information we collected, due to which we were not able to directly identify the impact of external factors on responses to collaborative scaffolding. In future studies, we will gather additional information concerning socio-economic status and pre-existing inter-player relationships, as well as proficiency in collaborative skills. From this data, we will be able to compare the impact of scaffolding on students of different backgrounds, different academic and collaborative skill aptitudes, and in groups with different dynamics. We will also be able to study the interplay of socio-economic factors and inter-player relations.

To get more significant results with this supplementary information, we will run future trials with more diverse demographics and environments. Our current study is limited to middle class middle school students in the classroom, but we would like to expand our work to both younger children in an informal setting and adults in the workplace. We will also explore different types of institutions, such as all-boys and all-girls schools, as well as alternatives in class size and teaching style.

Moreover, we plan to alter elements of our study to explore their impact on the effectiveness of collaborative scaffolding. For the video game, we will vary the genre and educational topic, while the scaffolding will change to promote different aspects of collaboration from our taxonomy. We will also vary our timeframe to investigate differences in short and long term effects, potentially performing an intervention over a series of sessions.

8 Conclusion

Thanks to advances in technology over the years, people now collaborate not just locally but across the globe through a wide variety of physical and digital mediums. However, because there is no proven optimal methodology, collaboration is often ineffectively taught. Our goal in this dissertation was to determine if collaborative scaffolding in an educational video game can promote the collaborative skills of performing part of a task for a group member (direct helping) and providing guidance to a group member (indirect helping). Along the way, we made five major research contributions related to collaboration, education, and video games.

We began our inquiry by clarify the term "collaboration". Through a literature review of over 25 studies and surveys from a wide range of domains and target demographics, we developed a definition suitable for collaboration in the context of collaborative scaffolding in educational video games. Thanks to this contribution, researchers interested in the topic can use our definition as a basis for their own work.

Because the definition is somewhat broad, we needed to deconstruct the concept of collaboration to narrow our research focus. Because we found no classification that adequately fit collaborative scaffolding in educational video games, we developed our own multilevel taxonomy based on patterns we identified in our literature review. We included example assessments and scaffolding statements to increase the utility of our taxonomy for researchers, educators, and developers. As our second contribution, this taxonomy can be

used to develop new evaluation techniques, explore novel research directions, and aid in classroom teaching and evaluation.

With our research question refined, we created a mixed-factorial study design to address our research goal and ran trials with local middle school students. While playing a collaborative educational video game, participants in the treatment group encountered collaborative scaffolding, while those in the control group saw neutral scaffolding. Before and after play, students also participated in a collaborative classroom activity we designed ourselves.

For the game, we used our own version of the Singapore-MIT GAMBIT Game Lab's *Waker*, in which we made several modifications including the implementation of a scaffolding architecture, our third contribution, and the creation of a data collection mechanism, our forth contribution. Researchers and developers can use our designs and our insights into impeding challenges to either build on our system or develop their own new infrastructures.

To detect instances of helping, we recorded audio and took observational notes. Data collected from within the game supplemented our understanding of player interactions and progress. We also had students fill out pre/post questionnaires on attitudes toward collaboration and knowledge of the educational topic our video game taught, as well as a questionnaire on motivation.

Our fifth contribution is the analysis and subsequent discussion of this data, which we presented in Chapters 5 and 6. Although we found no difference in helping behaviors among the control and treatment group, our questionnaires raised some arguably more intriguing questions. Collaborative scaffolding caused a decrease in *liking to collaborate* and *prosocial tendencies*, as well as lower *flow*, *immersion/presence*, and *engagement* scores. The latter three could potentially be explained by students pausing in-game actions to read collaborative scaffolding, but the others require additional research to uncover the impact of external factors like demographic and social dynamics.

In this dissertation, we developed a definition of collaboration appropriate for collaborative scaffolding in video games, presented a novel multi-level taxonomy of collaboration, designed scaffolding and data collection infrastructures, and uncovered unexpected reactions to collaborative scaffolding. Far from being an ineffective approach, collaborative scaffolding in educational video games has the potential to instill collaborative skills, but further inquiry is necessary to identify the factors that affect its impact on behaviors, as well as attitudes toward collaboration. By building on the theoretical contributions of this dissertation, we can start developing design principles and evaluation techniques to put truly effective collaborative scaffolding within educational video games in the hands of developers and educators.

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Appendix A Pretest: Collaboration

ID:____ Gender (Boy or Girl):_____ Age:_____

Collaboration Questionnaire #1

Circle the number that best matches how much you agree with each statement. The higher the number, the more you agree.

I like to share my ideas with others.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I try to share my materials with others when I think it will help them.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I like to help others learn.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

I believe that I can learn important things from others.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I try to help people who need it.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

Appendix B Pretest/Posttest: Education

ID:_____

Science Questionnaire

Draw a line from each term to its definition.

Term	**Definition**
Displacement	How far something moved.
Speed	Something with a value and a direction.
Vector	How fast something moved.
Position	Where something is.
Distance	How fast and in what direction something moved.
Velocity	The length of the shortest possible path between where something started and where it is now.

Circle the correct answer to fill in the blank.

1. Sam spends a whole day at home reading. As time goes by, Sam's **displacement** from home _____.

   a. increases
   b. stays the same
   c. decreases

2. Sam rides the bus to school. As the bus drives **to** school, Sam's **displacement** from school _____.

   a. increases
   b. stays the same
   c. decreases

3. After school, Sam rides the bus home. As the bus drives **away** from school, Sam's **displacement** from school _____.

   a. increases
   b. stays the same
   c. decreases

4. Jordan drives to work. When Jordan is **slowly backing** the car out of the driveway, Jordan's **velocity** is _____.

   a. big and negative
   b. small and negative
   c. zero
   d. small and positive
   e. big and positive

5. Jordan takes the highway to work, which means Jorden gets to drive **really fast**. While on the highway, Jordan's **velocity** is _____.

   a. big and negative
   b. small and negative
   c. zero
   d. small and positive
   e. big and positive

6. Jordan is almost to work, but has to **stop** at a red traffic light. While at the light, Jordan's **velocity** is _____.

   a. big and negative
   b. small and negative
   c. zero
   d. small and positive
   e. big and positive

7. Alex likes to ski. Alex rides a lift up to the top of the mountain. The lift always goes the **same slow speed**. As the lift moves, Alex's **velocity** _____.

   a. gets smaller
   b. stays the same
   c. gets bigger

8. Alex takes a lift up and then goes down the mountain. As Alex skis, Alex goes **faster and faster**, and Alex's **displacement** from the **bottom** of the mountain _____.

   a. quickly gets smaller
   b. slowly gets smaller
   c. stays the same
   d. slowly gets bigger
   e. quickly gets bigger

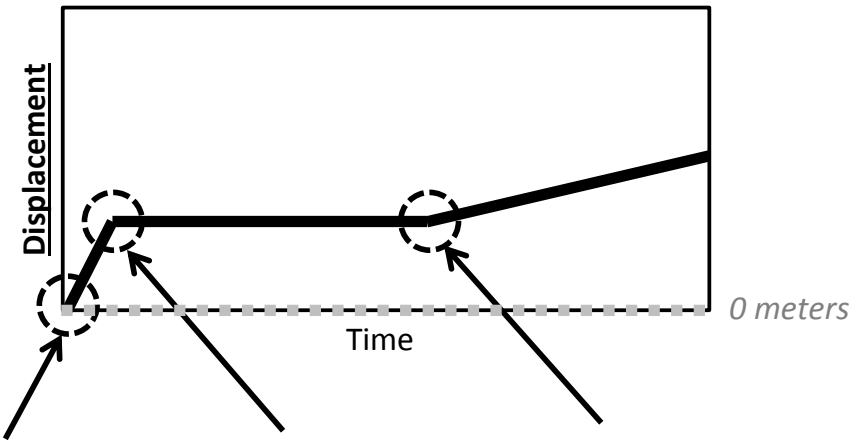
9. Alex takes a ski lift up and then goes down the mountain. As Alex skis, Alex goes **faster and faster** and Alex's **velocity** _____.

   a. gets smaller
   b. stays the same
   c. gets bigger

Use the phrases in the bank to describe what happened based on each graph. You may use phrases more than once.

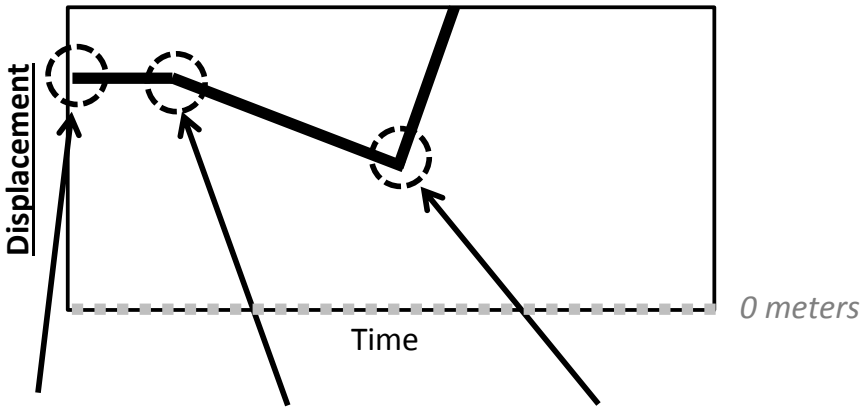
Started moving away slowly	Started moving away quickly
Stopped	
Started moving closer slowly	Started moving closer quickly

1.

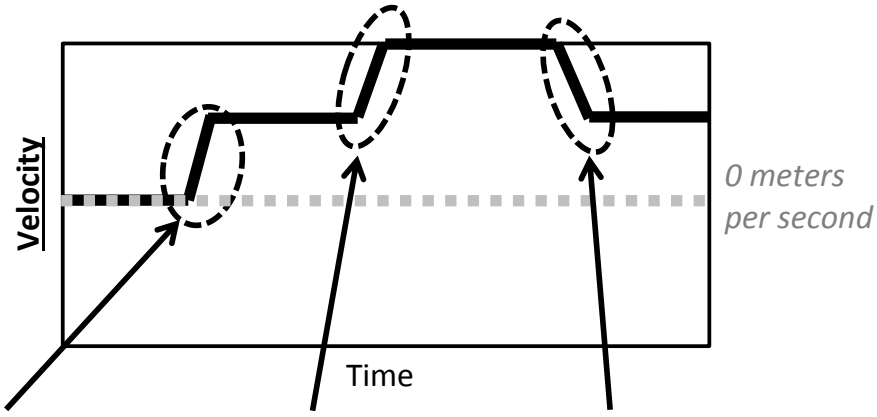


_____ _____ _____

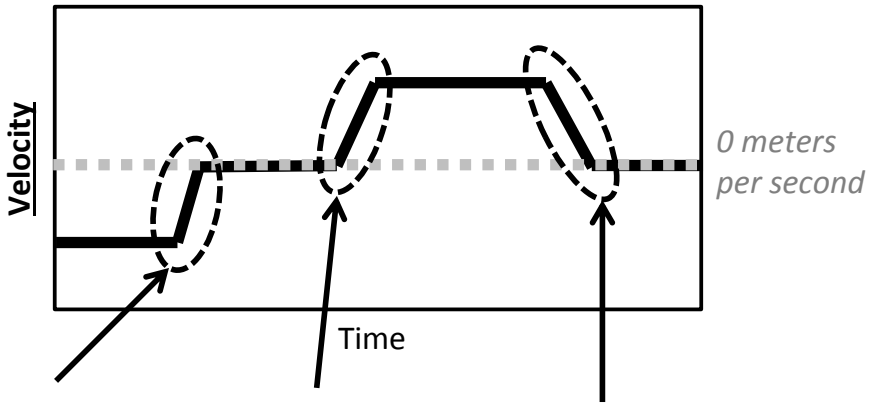
2.


Displacement
0 meters
Time

3.


Velocity
0 meters
per second
Time

4.



Appendix C Motivation and Engagement

ID:_____

Motivation Questionnaire

Circle the number that best matches how much you agree with each statement. The higher the number, the more you agree.

I lost track of time.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I was tired from playing,
but only noticed after I was told to stop.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I felt like I was actually in the game world.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

I put a lot of effort into playing the game.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

My interactions in the game seemed real.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I tried very hard to win the game.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

When someone talked to me,
I didn't notice because I was focused on the game.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

I cared about the outcome of the game.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I lost track of my real surroundings.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I enjoyed playing the game.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

Appendix D Posttest: Collaboration

ID:____

Collaboration Questionnaire #2

Circle the number that best matches how much you agree with each statement. The higher the number, the more you agree.

I like to share my materials with others.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I believe that people should help each other learn.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I don't hesitate to help others when they ask for it.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

I like to cooperate with others.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

---

I try to share my ideas with others when I think it will help them.

1	2	3	4	5	6	7
Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree

Appendix E Scaffolding Architecture

Table E.1: Scaffolding statements and trigger events

Level	Target Player	Event Type	Event	Collaborative Scaffolding (Treatment)	Neutral Scaffolding (Control)
1	0	Inactivity	Does not pick up an orb within 10 seconds of level start.	"I should create a path for my group."	Selected from Table E.2.
	1	Inactivity	Does not pick up an orb within 20 seconds of level start.	"I should help create the path."	Selected from Table E.2.
	1	Hazard	Cumulatively, players hit hazards 4 times.	"Discussing how to get past these obstacles might not be a bad idea."	Selected from Table E.2.
2	0	Location	Reaches exit door but partner does not within 5 seconds.	"I think my partner is going to need my help, too."	Selected from Table E.2.
	0	Hazard	Partner hits hazards 3 times.	"Redrawing my partner's path to make it easier may not be such a bad idea."	Selected from Table E.2.
	1	Location	Reaches exit door but partner does not within 5 seconds.	"I wonder if there's anything I can do to help my partner like they helped me."	Selected from Table E.2.
	1	Hazard	Partner hits hazards 3 times.	"Maybe I should redraw the path to make it easier for my partner."	Selected from Table E.2.
3	0	Location	Tries to reach area only accessible with partner's help.	"I may need my partner's assistance to get further up."	Selected from Table E.2.
4	0	Location	Tries and fails 3 times in a row to take harder path, which does not require partner.	"It may be easier to get across if we work together."	Selected from Table E.2.

Table E.1: Scaffolding statements and trigger events (Cont.)

Level	Target Player	Event Type	Event	Collaborative Scaffolding (Treatment)	Neutral Scaffolding (Control)
4	1	Location	Tries and fails 3 times in a row to take harder path, which does not require partner.	"Maybe if we help each other, we can reach the exit easier."	Selected from Table E.2.
5	0	Location	Reaches exit door but partner does not within 120 seconds.	"I should probably help my partner by redoing my path."	Selected from Table E.2.
	1	Location	Spends 10 seconds near orb that partner needs to pick up.	"I think I'll need my partner to pick this one up."	Selected from Table E.2.
	1	Location	Reaches exit door but partner does not within 120 seconds.	"I may have to redo my path so that it's easier for my partner to get across."	Selected from Table E.2.
6	0	Location	Tries and fails 3 times to take harder path, which does not require partner.	"Maybe there's an easier path up if we work together."	Selected from Table E.2.
	0	Location	Partner tries and fails 3 times to take harder path, which does not require the player's help.	"I wonder if there's some way I could help my partner."	Selected from Table E.2.
	1	Hazard	Partner hits hazards 3 times.	"Perhaps I should redo the path to make it easier on my partner."	Selected from Table E.2.
	1	Inactivity	Neither partner progresses through the level to a certain location within 120 seconds.	"I should probably consult my partner to help me make this graph."	Selected from Table E.2.

Table E.1: Scaffolding statements and trigger events (Cont.)

Level	Target Player	Event Type	Event	Collaborative Scaffolding (Treatment)	Neutral Scaffolding (Control)
7	0	Orb Take	Partner picks up an orb, but player does not activate switch within 60 seconds.	"Is there some way I could help my partner make the path higher?"	Selected from Table E.2.
	1	Orb Take	Partner picks up an orb, but neither player effectively uses the resultant path within 60 seconds.	"I wonder if I can help my partner figure out how to draw the path."	Selected from Table E.2.

Table E.2: Neutral scaffolding statements

ID	Statement
0	“These ledges are slippery.”
1	“I sure am fast.”
2	“I hope it doesn’t start to rain.”
3	“I like clouds.”
4	“It’s kind of cold out here.”
5	“The color blue is pretty.”
6	“I wish I could fly.”
7	“I’m happy.”
8	“Water tastes funny.”
9	“This place needs more trees.”